

2nd Applied Active Inference Symposium on "Robotics" ~ 2nd session

Applied Active Inference Symposium | 2nd session | 4:00:19

All right.

Hello and welcome, everyone. Welcome back to the second Applied Active Inference Symposium.

It is the second interval, and it's July 31st, 2022.

In this second session, we are going to first feature a presentation by Bruno Lara, Prediction Error Dynamics, a proof of concept implementation.

Following that, we'll hear from Matt Brown with a presentation, Real-Time Robotic Control Through Embodied Homeostatic Feedback.

The third presentation will be by Adam Saffron on Generalized Simultaneous Localization and Mapping, GSLAM, as a unification framework for natural and artificial intelligences.

And the final presentation will be by JF Klotjer, Towards a Symbolic Implementation of Active Inference for LEGO Robots.

We'll then have some of those presenters rejoin us in the final two or so hours of this interval for a roundtable discussion.

We'll also be joined by Carl Pristen there.

So thanks, everyone, for watching live or in replay.

Hope that you add any comments, if you would like, into the live chat.

And otherwise, thanks to all of our presenters and co-organizers for their awesome work.

And thanks so much for joining, Bruno.

Please take it away.

Yeah, thanks a lot, Daniel, for the invitation.

I was very surprised, as most of our work so far is not directly related.

Well, it's not actually Active Inference.

Of course, it's related.

But since we published a couple of years ago, a paper that actually I'm presenting a bit of these results here.

Well, lots of nice and interesting people started to look into this implementation of predictor aerodynamics.

And it's actually the part where we could say it's the part that we are taking out of all this framework to try to implement in our robots, in our agents.

I don't know how much or how many times you've heard this story of why we are doing this kind of cognitive robotics

or why we like to call all this research or this area cognitive robotics.

And I like a lot this slide, which kind of summarizes what it's usually a long introduction on the history of artificial intelligence and robotics.

And it just tells this story of 50 years of research trying to figure out how can we have machines that are as intelligent as humans.

And it comes out that actually this upper part of the image quite got resolved rather fast.

We cannot say easily, but it was more or less fast.

And then we have the second part of the slide, like the bottom part, which is what comes out to be the most difficult challenge for robotics.

So what has to do with interaction with the world, you have a small child that is capable of handling these objects in the environment.

And then you have this other robot that can do it.

But at the moment that you change the size, the weight, the position, whatever you want of the pieces or of the chessboard,

you actually have a child that with no problem can adapt himself to the challenge.

And the other joke is then on the other side, you have another PhD that has to solve it if you do the same for the robot.

So in reality, what was supposed to or was thought to be kind of the hard problem, which was having a robot reason and solve problems,

it came out to be quite not so difficult to solve tasks.

And we can talk about more difficult challenges that have already been passed, such as, I don't know, playing Go, which is way harder than chess.

And we had AlphaGo a couple of years ago, well, winning against the human champion.

So we like to think that we are doing now things differently.

We are having robots that interact with the world, that interact with what's going on around, and they learn through this interaction.

And of course, that gives us lots of challenges and lots of things that we have to solve.

So now I come from this, yeah, you could think now as an old school of internal models.

We work in this field for quite already some years.

And it's this part of the world where there is inverse and forward models for control, of course.

And they are thought to be behind all these important issues in humans.

Since the beginning of my career, we've been doing implementations of different types of tools for these robots,

but always based on these inverse and forward models.

So just a quick reminder for those that might not be very, very much into this type of models.

So we had a sensory situation, whichever it is, whatever it is.

And we have a desired sensory situation.

We have an inverse model that takes us input those two things, the sensory situation at time t , of course, and the sensory situation at time t plus one, which is the desired.

And the inverse model is going to tell us what's the motor command that needs to be performed to go from here to here.

And then we have a forward model that takes the sensory situation at time t and the motor command at time t and gives us the prediction of what's going to happen with this sensory situation.

And so we have a prediction error when we, well, we have lots of errors,

but we are interested on particularly this error, which is the difference between what we actually predicted and what happened in the world.

So we've been doing this for quite a while.

So, well, we've also used this prediction back as a sensory situation, and we can get long-time predictions, what we called, we can do lots of comparisons between what we wanted and what actually happened, and lots of different errors.

But as you know, and well, now everyone works with, is actually this error.

Like the difference between the world and what actually I predicted.

Yeah, and so that's what we can say we have in common.

So some years after, well, some years ago, actually, we came across all these ideas of prediction, sorry, predictive processing, and we started to dig into it to see what actually we could use for our models, what could be useful for improving our models and our agent's behaviors.

And we have lots of doubts, lots of questions.

But one of the first things that we've done so far is to look into what happens with the error as an agent is executing an action.

Sorry.

But first, I'd like you to walk you through one of our implementations.

Well, first, where is actually the, what you could call the cross areas of what we believe it's important in this case,

when we have active inference and prediction error minimization.

So most of us are very familiar with this kind of representation of what's happening actually during action.

And we actually believe that prediction, minimizing prediction error is what leads to belief updating.

So like the change in a prior belief encoded as a posterior belief, and this is actually informational gain.

So this means learning, right?

So in active inference, the expected future states are fulfilled through action execution, we know.

And this associated expected prediction error is minimized by actively sampling sensory information.

And so the important part for us is this expected prediction error.

And we believe the attribute of a policy, which can be epistemic and instrumental, in case this action is novelty searching,

it's reflecting its epistemic affordance, and we could say it's acting for information.

In case of preference searching, no, this reflects instrumental affordances and its action for reward.

So in words of Kirchner, we can think of active inference as the process of selecting relevant affordances.

So those which are expected to minimize prediction error in a context sensitive manner.

Yeah.

So what we've done actually, or what we wanted to actually implement is

how these expectations about the dynamics of the error cause or are related to feelings, to these embodied feelings.

And we believe they allow sensitivity.

So how well or how bad an agent is doing at improving the grip on what is relevant in the landscape of affordances.

So the sensitivity to changes in the rate of prediction error reduction is what we call prediction error dynamics. And we can say the steeper the slope, the faster the rate of reduction. And we see some examples, just very simple. So in red, in black, we could see the actual error as the action is executing. And then we have a slope associated to this error. So what we don't want, what the agent doesn't want is this error increasing, having a positive slope. And on the other hand, the error is decreasing. Then we have this slope. And this is actually what we are looking for. So prediction errors are manifested as changes on affect. So if the rate of prediction error is faster than expected, we have a positive balance. And we have an action policy which is more precise. In the case that the error is slower than expected or is increasing unexpectedly, we have a negative balance and an action policy which is less precise. So just walking you through a couple of implementations that we have done. The first one is just kind of an example of this mix that we are trying to have. The error is used for learning, but we have a very simple implementation of self-organizing maps. We call it the self-organized internal models architecture. The nice thing on this architecture is that it actually sort of doesn't have to have an inverse model. What we have is a self-organizing map that organizes the motor sensory information. And on the other hand, we have visual sensory information, which is also organized in the same another self-organizing map. And then we have what we call a multimodal representation. And in this multimodal representation, we have associations. We have associations between a sensory situation, another sensory situation, and a motor command. So you could think of this as an inverse forward model pair, but it's actually not encoded as such. So we just have associations between sensory situations and motor commands. So we thought this was very interesting. And we use it, for example, for saccadic control. So the robot has some initial sensory situation, which is the stimulus somewhere in its field of view. And what we need or what the robot wants is to fobate the stimulus. So this is the sensory situation, the desired one. And we ask this self-organizing maps architecture what we need to do to bring the stimulus from here to here. And it's executed. It's not very precise. So it comes close to the center of the image. And then we use this as the new sensory situation. Then we have the same target.

We again ask the architecture what do we need to do?

And then it comes very close.

So after a hundred runs, you can see that the robot, this is the starting situation.

In red, we have the first prediction and execution.

And in green, we have the second prediction and execution.

So it's a very nice, simple architecture that actually works.

As I said before, it works like a forward model, but it can also work as an inverse model.

We use prediction error to learn.

And so we could use it as some kind of modular architecture.

We can put more maps.

And in this case, we have the position of the head, visual stimulus, the position of the arm.

And what the robot can actually do is find the stimulus somewhere in the space and then bring it to the center, kind of obviate it.

And then actually bring the arm also there.

So we have all these associations between movements of the head, movements of the arm, movements of the eyes.

And all these get associated and are coded as initial and ending positions.

Okay.

So the next question or the next thing we tried to do was how can we use prediction error dynamics to guide learning?

And this was this paper we presented a couple of years ago in Epiroc.

It was actually Guido's that presented it, which is our co-author.

So it's a work by Guido, Alejandra Asiria, which is also a colleague and myself.

And what we did there is.

I won't work on the on the I won't tell you much about the the.

Max or the or the kind of networks that were used in detail.

The important part is what we see here.

So we have a very simple robot.

This is what you could see here.

It's a two degrees of freedom camera mounted on on a sort of artificial field where we have lots of lots of plants.

And the the the robot moves towards a specific destination.

So it looks it wants to look at some at some image at some plant.

And so we have for each movement for each goal.

Sorry for each goal.

We have a prediction error here.

So we organize the goals in a zone.

Of course, the goals sorry are encoded using an auto encoder.

We use the middle part of the encoder to.

Have represented the goals here.

And the important part is each each one of these goals has some prediction error dynamics.

Yeah.

And so.

We go back again to our inverse model and forward model and execute the command, the motor command, and we get.

And we get.

And we do our prediction and we get an error between what we wanted and what happened.

And so.

The interesting part and what became kind of nice and what that was what we wanted to implement.

What we were thinking about this.

We append this prediction error to us a sort of a buffer.

And this buffer.

Once that it contains more than four predictions, we do a linear regression.

And actually what we get is an indicator of the progress.

How good are we doing?

So it's kind of a tendency for, for, for, for learning.

So when we have a negative slope, it actually means we are going towards the goal.

We are doing fine.

So progress is increasing.

And it means there is a positive emotion and we remain on this goal.

So learning is going fine when the, when the slope is positive.

So the activity is away from the goal.

There is no progress.

The error keeps increasing.

So we have what we could see as a negative emotion.

And something that can be encouraged is then to switch, to switch goal.

So we have a positive slope means increase the size of the prediction error dynamics buffer.

This was one of the, sorry, that was something of kind of interesting and, and, and very, yeah, new in, in, in, in, in this, in this field, in this area.

We haven't thought about it before.

And what we actually did is, okay.

What happens if, if my, if my slope is positive, it means I'm doing very bad at this task.

And what I'm going to do is I'm going to increase the size of this buffer.

Yeah.

So when, when, when we have an, a negative slope, it means, okay, I'm doing fine.

So I'm going to decrease the size of this buffer.

So what we are actually doing is adjusting dynamically the size of this monitoring window.

And, and, and, and we're doing that depending on how things are going with respect to what we are actually expecting.

So if the prediction error increases, we have a positive slope and we do a more frequent evaluation of the feedback.

So, this allows a quick correction of the action.

And it can lead to satisfaction in case where we, we, we, we managed to, to, to do it better and induces abandoning the goal.

So after, in this case, after a number of interactions, interactions, where the error is not decreasing, the error keeps increasing, then we can change, we can switch goals.

And we could say we are trying to avoid frustration.

So through intrinsic motivation, what we try is search for a new goal.

When the prediction error decreases, we have a negative slope.

We don't need to, to be evaluating the feedback so, so frequently, and we can free resources for the agent to do something else.

There is no more prediction error.

So that means we've achieved our goal.

So learning happened, learning occurred.

Again, we search for a new goal now.

So when we select a new goal, we give priority to those that have a negative slope, which is kind of steep in the prediction error dynamics.

And we have some examples here of how the error moves as we are moving in, in, in, in the world and how the buffer size increases when the error increases.

And then it decreases, how, when, when the slope now again changes.

So that was, that was one of the first things we did.

And now actually Adam invited us to this, to this special issue that he did in Frontiers.

And, um, we, we published there a paper, which is actually, well, at the beginning, it was about something else.

And then we ended up with this, with this, with this model presenting this model.

Uh, this is, this is just, uh, an idea that we have, which actually encloses this part that I talk about, about the slopes and the expected error.

So what, what we are, what we are suggesting here is that, um, I don't know how much time we have, if I can go into detail on this, but, um, let's, let's try something very fast.

So we have an agent that has maybe physiological needs in case the agent doesn't have psychological needs through intrinsic motivation.

We want to do something new.

We want to do search around the world.

Then we go to what we call the environmental context, which brings us to this field of affordances, uh, where there is lots of actions and each action, each, each task has some, uh, expected error dynamics.

We got to a task related context, which comes from selecting the task.

And then we do some planning of sensory motor sequences.

We start acting on the world and we get some error.

As we execute this task, then we have two types of monitoring.

We have the typical, um, momentaneous say error that helps us go through the planning and correct the, the execution of the task.

And we have a, a, a window of prediction error dynamics, which is compared to the expected error that, that this section had.

And according to that.

Is that we are going to go to these emotions.

That tells us, no, the, the error is increasing.

We have this slope here.

So we go back to another task.

Uh, before like, of course, checking then if something has changed, we still have physiological needs or we don't.

If the error is not so gravely increasing, it's more or less going fine.

Then we go back to this same, uh, planning to the same action.

So it's, it's a very, uh, very, very nice, uh, model that we thought of it's, it's, it's, it's, it's on the, it's on the make.

Um, and we have many ideas, uh, from bringing this box diagram into, into the world.

And, and in our, in our idealistic, uh, um, dream, what we are thinking on is something actually based on SOIMA.

We haven't implemented this, uh, yet, but it's something based on this SOIMA.

But now we have lots of other modules.

We have, uh, some interoceptive, uh, internal model representation, exteroceptive, um, model representation.

Uh, this is actually wrong, but anyway, so in this side, we have all the exteroceptive information, the interoceptive information, and we have some policies and a task and some error.

This error is going to tell us how we are doing, how we can stay on the task.

Uh, if the error is too big, then we can change tasks.

So it's a bit of a mixture between the, the, the previous thing I told you, and, and this is how it could look in a, in a, what happens on a top down.

So when we have a task, this task is going to activate some policies and these policies are going down, um, to act on the world.

We have again, again, uh, some error coming up and this error is somehow, somehow coded here.

There is some monitoring so that we can go on in our policies.

This is the momentaneous error.

If the error is very big, then we can go back here.

Um, ah, sorry.

So this is top down and, and this is actually bottom up.

No, it's actually the error coming up all the way and coding here.

And, uh, yeah.

And this is how we start the cycle with some task policies and everything else that is happening in the world.

So I think that we can, we can, we can stop there.

I hope it's, it's clear.

We don't have direct questions from the audience as I'm not seeing anyone or.

I'll ask you one question.

Otherwise, no, we have no direct questions, but we can, um, definitely accumulate them for the round table.

Yeah.

What is the relationship between prediction error minimization and free energy?

Um, well, I, I think it's just like part of the process.

Uh, I, I, I don't know what you mean.

I mean, within the whole thing of, of whatever you want, free energy, um, everything that is happening, the difference between the expected, um, states and whatever.

Actually happening in the world.

So we always have in, in, in active inference or in free energy, you always have an error happening in, in, in what's happening.

What's in what's.

Yeah.

What is happening in the, in the action of the, in the world.

So through this prediction error is how you are.

We, we believe it's going to happen.

Like the learning is going to occur.

I don't know.

I don't know if.

Yes.

That's awesome.

Well.

Yeah.

Yeah.

Thank you very much.

Hope to see you very soon.

Yeah.

We'll see you like in, in two hours or so.

Excellent.

See you soon.

Yeah.

Okay.

Thank you very much.

Hope to see you very soon.

Yeah.

Thank you very much.

Hope to see you very soon.

Yeah.

Yeah.

Yeah.

Yeah.

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Thank you.

a 20-node homeostat. It takes a bit longer, and it's a little bit more demonstrative. So let me switch briefly here and apologize for the delay while I get this up.

Cool. All right. So let me just describe what you're seeing here. You're seeing 20 single-dimensional graphs. This is the accumulated local prediction error at each node of the 20-node homeostat. You can also think of this as one 20-dimensional dynamical tractor or 20 single dimensional dynamical tractors. These are also kind of, you consider these as phase-based graphs for each of the nodes. So what you're seeing here is I have it paused. I'm going to move it along slowly. This is slowed down on purpose. And what you'll see is they start off very chaotically. And what will happen is a couple of nodes will start stabilizing, then more will start stabilizing, and then eventually the entire network will converge to zero prediction error. Now, this is unconnected to an environment, so it's just predicting itself. One thing I like to do here,

and I don't know how many people else are going to find this interesting. I want to show a little in-depth look at what this final convergence and collapse to zero looks like. So what I'm going to do is I'm going to change this to auto-scaling the graphs, meaning that as it goes to zero, you'll see the graph ranges drop. And what you should see is usually there's a pattern forms amongst the output nodes that is kind of consistent with every drop. What you'll see is that, you know, it's kind of hard to see with 20 nodes, especially I've gone a little too far. We're already at E negative 20 here. But what you're seeing is you're seeing the 20 dimensional drop to zero, and there's kind of like a cyclical pattern to this. And every time it goes to this pattern, it drops down another order of magnitude. And this will go all the way down to the floating point error of my computer, which is about negative 46, I believe. And then it basically fails to function at that point because it can't tell the difference, floating point error, it can no longer function. So, and the whole thing kind of falls apart there. But that kind of shows you a bit about the process. Let me go quickly back to the slides. So yeah, that just kind of showed you a slowed down version of the 20 dimensional homeostat. And I wanted to show the 20 dimensional because it happens a lot slower, and it's a little easier to see what's going on with four nodes, it's almost instantaneous, and it's hard to see the progression. Okay. So that seems like a lot of work for a model that doesn't really do anything and isn't connected to the environment. So, you know, we have this model that can adapt this internal structure to reach homeostatic equilibrium through kind of a self synchronization process. How do we use this to drive behavior? And you know, this next step took me a few years, even though it's kind of conceptually simple. But don't worry, it'll get complex really quickly. The short version is sensors and motors kind of define how the agent is embedded into its environment. And the network topology defines the adaptive capacity of the homeostatic agents model. Sensor definitions provide a mechanism to influence the behavior of the system by providing priors in the form of preferred environmental states. These act as kind of like TLA dynamic priors, externally defined dynamic set points for the homeostatic adaptation. The motor definitions provide critical affordances for modifying the environment that provides the basis for future sensory predictions. The preferred states of the sensors become the preferred outcomes of the motors, and the model's behavior is biased as desired. This is probably most people on this active inference symposium are probably familiar with this idea. This slide is put together for the benefit of people who are kind of familiar with traditional deep learning and neural network approaches to contrast how these networks function differently, because it's a bit hard for people to get their head around. The first is probably the most obvious is there's no back propagation here, only ultra stability, which is kind of a local rule. This leads to huge sample efficiency and the learning is very structured, meaning you don't have to, you know, diffuse learning across, you know, thousands or millions of parameters, you know, you only have to model the part of the, you only have to update the part of the model that is mispredicting or, you know, is part of a misprediction. The other thing to notice is that this is not atemporal. Most neural networks and, you know, like deep learning networks are kind of disembodied atemporal processes. This is very much a temporal

process. You know, signals take time to move from one node to another, which makes it inherently the whole thing very temporal. The resulting temporal relationships are important, actually. Spatial relationships of sensors and motors end up getting encoded into temporal, relative temporal relationships within sensory signals, such and also vice versa. Signal timing timings of the propagating signals within the network can translate into different physical spatial relationships amongst the output motors, for example, or the input sensors. So there's a relationship between temporal location, relative timing of signals and the spatial location of sensors and motors and the physical embodied agent. Different in the, you know, just to kind of, you know, push on that a little farther, the different regions within this larger homeostatic network will then be sensitive to different sets of motors and sets of sensors based on their physical relative location. And by contrast, you know, you know, deep learning, deep reinforcement learning, deep reinforcement learning, for example, is an attempt at adding temporality to an otherwise atemporal static deep learning models. You know, the other way to compare these two is, you know, deep learning models are generally trying to be general function, universal function approximators, and where this is much more generative and inherent to temporal, the very learning is very, very different. And just to expand a bit more on the topics from the previous slide, the network diagram here on the right is very oversimplified network, and it's just meant to visually explain how to think about some of these networks. There's no forward or backward, no front, back, top, bottom, only kind of relative location from other nodes and from sensors and motors, perhaps more relevantly. And it is, you know, common for people to look at the learning process as a sequential flow of processing, like sensing, perceiving, then acting, or planning, then acting. And instead, you know, you have to kind of think of this as all happening at the same time, merging together as kind of one whole perceptual motor system, not a traditional sandwich model. Time simply advances and we simulate the analog systems as signals propagate across the network. And so, you know, you don't get input from the sensors, calculate the network and get out from the motors, you get you send input to the sensors, you get out from the motors based and those motor actions are basically based on the generative models prediction of what it should be doing already before having even sensed the world. So the, you know, what's interesting here from the model point of view in terms of how this homeostasis works is the network's homeostatic feedback loops come to be dependent on the expected sensory signals, meaning that the sensory signals become entrained in the homeostatic self-maintenance processes, which kind of naturally makes it grounded in the embodied experience. And this, you know, homeostatic self-maintenance process is composed of the environmental signals themselves as part of its predictions and expectations. Expectations and predictions for the consequences of motor actions are embedded in the signals sent it to the sensors from deeper in the network, like on the right here. And they're sent to the sensors where expectation meets reality and prediction errors cascade backwards and modify the internal structure of the model as polygraph's model update. And then motor actions are accompanied by these parallel sensory prediction signals that flow from the deeper in the network. The actions taken at any moment are not based on the

latest input, but based on predictions driven from deeper in the network.

David Plylariis

This may be familiar to those who are familiar with predictive processing or perceptual control theory. And the idea here is that where the sensors and motors meet the network, the network itself is trying to make those local prediction errors zero, meaning it's trying to anticipate the actual sensory experience through kind of inverse modeling.

David Plylariis

David Plylariis

This process can be seen as generating an adaptive synchronization manifold between an agent and environment around minimization of prediction error. This is kind of looking at synchronization as prediction, you know, when, when a system, when two systems are synchronized, they can be thought of as predicting each other anticipating the other, the other's motions and individual nodes using ultra stability are using ultra stability to learn how to predict their neighbors and when at equilibrium these nodes are kind of synchronized through these feedback loops and the synchronization spreads from individual nodes to whole network synchronization and ultimately uh out from the motors and back and through the sensors um in a synchronization manifold between the agent and environment um you know to say it another way the the overlapping feedback loops interact and synchronize with each other as part of this homeostatic self-maintenance and as these feedback loops extend from motors through the environment to the sensors the model gains the ability to anticipate the consequences of its own actions uh and the capacity to synchronize internal states with something in the external world the synchronization means that you know internal states will come to resemble uh the uh one will come to resemble the internal states of the other um this is kind of along the lines of you know the good regulator theorem um when at this you know synchronized homeostatic state it's also in this critical state i mentioned this earlier um where it's stable um but it's sensitive to local prediction error so the meaning that the the model uh this critical state may is maintained as long as local prediction error is maintained within a certain threshold um and but otherwise you know the smallest prediction error can trigger global kind of structural changes within these uh feedback loops uh in order to better predict the future uh you know in order to maintain a synchronization network must be able to converge to equilibrium faster than the environment can destabilize it um and so there's um you know so there's update you know network update rates where we have to maintain um you know this kind of fast movement and this kind of is in line with the law of requisite variety where we have to make sure that we can um explore state spaces faster than the environment we're attempting to control um and as long as the homeostatic network can re-establish establish these stable feedback loops faster than this synchronization manifold will be maintained uh and synchronization is kind of an interesting thing um especially in the context of robotic control um because you know once kind of two elements are synchronized you know let's say you have a sub network that's synchronous synchronized and you know two indirectly two nodes that are not directly connected uh can end up coordinating without any kind of communication between the two because they've been previously

synchronized uh their behavioral end up being correlated as well um

and in case it's uh interesting to think about um you know what is the structure of these kind of feedback loops and this you know generative model and the way i like to think about it is a dynamic holarchy meaning um you know a hierarchy is kind of like uh you know a top-down construction um but you know a dynamic hierarchy is more like a um a bottom-up construction where the parents are composed of the children and dynamic in that as the children change their own self-definitions the parents are likewise um changed so it's kind of constantly shifting and changing over time as the children redefine themselves uh and this structure is very useful uh as a gender model as well as a small number of individuals to you know kind of reshape the hierarchy very quickly as it's part of an ongoing dynamic process um cool so i'll switch over and do some quick kind of uh demonstrations showing how this ends up you know why would you actually want to do this um you know yeah why would you actually want to do this especially in the context of robotics um so the first thing i'm going to show is um let me just share here quickly this is the demo uh i put together um a couple years ago actually let me let me shrink this screen a little bit so it's not quite so much to share

uh this was put together a couple years ago to kind of demonstrate you know why would you do why would you do robotics like this you know what is the point um and what is robustness mean and what does adaptivity mean uh in the context of uh robotics so hopefully you can see now uh my screen this is just a uh simple um modification of uh an open ai gym called uh reacher uh and what it's been modified to add a single texture joint here uh and the reasons will be obvious here in a minute um but um this is just a simple task it's try the you know the model has to get its end effector to the target position uh and this is using these homeostatic models here um but here's what's interesting i can do and it hasn't this model hasn't seen this situation i can disable the outer motor for example and it will basically on the fly figure out how to achieve the goal um you know just by updating its internal model and saying oh this motor doesn't work anymore let me let me modify my behavior my model of the world i can turn that back on and um he'll start using it immediately or it will start using immediately i can do the same with the internal motor um it can turn that off and so oops i did not turn it off there we go now now it's off um yeah so and it just kind of figures out oh okay the the inner motor i have no longer have access to let me update my model and i can again turn that back on and it'll start using it again um so this was just kind of a simple demonstration i made a couple years ago to show why you would want to do robotics this way in that you can kind of get this real-time adaptation to unexpected situations um due to the the gender model that's constructed this way um now let me just scoot forward a little bit to more recent work um and hopefully um the frame rate won't be too bad over zoom um let me just shrink the window a little bit in hopes that helps

okay let me um share this

the frame rate is feels a little rough over over zoom here um but this is all just running locally on machine um so this is something a little more uh recent uh a little more complicated we made the goal a little bit more compli uh complex uh and we've used a full six degree of freedom arm here from like an uh this is actually a fetch robot uh mobile arm um but something a little more realistic than

like a toy problem uh and i'm just you know continue you know periodically here retargeting the the desired angle for this uh valve and so this is a robot that it that turns valves uh and what i've done is that um i i can periodically change the set point for its you know where how it sees the world basically you define goals by you know defining preferred states of the world uh and then the homeostatic model basically figures out how to control the robot to it to uh to affect the you know the state of the world that it wants um i'm hoping here every 30 seconds it will randomize the um the valve shape the square one here is the simplest one but let me add some random motion and random orientation the model has not seen this before either so it just kind of learns on the fly how to deal with this kind of moving shape as i do it i can also do the sinusoidal motion here so imagine it's on a floating platform um where the rotation and the uh position is kind of moving through space um here we go now we got something more interesting so now we have it still moving through space it's a circular valve here something a little harder to turn um and what i'm going to do is i'm going to make it very slippery so imagine i've just poured oil all over it so now it's having to deal with a really really slippery valve here that it has to turn and it's still manages to do it even though it's really hard um but you can get the idea here is you know uh if you do the generative model in this kind of sample efficient way you can kind of um get natural um you know adaptation to whatever it's kind of experiencing in the moment here um so i think i've kind of i think this is maybe i maybe i'll do the random motion again maybe that's interesting um but yeah there's all sorts of stuff you can kind of throw at this and it'll kind of deal with it uh apologize for the frame right i'm going to blame zoom for that um so uh cool all right good um i don't think i have too much left um but you know uh uh next steps are uh basically improving automation and tools um i'm working uh my startup is basically developing this for um for real world use uh in adaptive situations where you need um fine motor control uh and kind of uh dexterous object manipulation in uncertain environments um you know it's on the technical side we're looking at you know scaling up goal specifications and topologies which are related as well as you know deeper temporal horizons uh in both directions planning uh and uh head and recall um but yeah that's um about all i had for today and and i look forward to the questions during the uh the um round table in a little bit if anybody has questions now i can uh talk to it as well i'm a little i went a little fast i apologize i'm a fast talker it was great just maybe one preliminary question what settings do you see these uh kinds of implementations occurring in first and then the other question in order or if it's your preference is how do you see the homeostatic network related to the analytical first principles foundations of active inference as we know it today yeah no um those are really good questions and i you know i i didn't talk much about active members because i figured uh everybody else had kind of done it pretty thoroughly um the use cases first your first question um use cases we're looking at are things that are just beyond um the ability of uh current robotics um the idea here is this software could be put on you know any old robot and kind of provide you know uh adaptive capabilities for certain things um you know we're pretty open-ended in terms of what we're looking at in terms of use cases uh right now our current uses are in like um uh energy uh and manufacturing uh particularly um this this is a fair amount of

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All right. Welcome back, everyone. We are continuing on with Adam Saffron, Generalized Simultaneous Localization and Mapping, GSLAM, as a unification framework for natural and artificial intelligences. So, Adam, thanks a lot for joining, and please take it away.

And today I will be discussing a grand unified theory. Another one.

I recently co-organized a special issue of *Frontiers in Neuro Robotics* with Ines Hippolito and Andy Clark.

There's also another special issue that I am co-organizing with other active inferencers and Fristonians for Royal Society Interface Focus on the subject of symmetries in mind and life broadly construed, ranging from symmetry breaking and dynamical systems to a gauge theoretic formulation of the free energy principle.

So, in this work and throughout all of it, my journey, I've been trying to understand the bases of autonomy in biological and artificial systems.

These are the kinds of questions that have motivated me.

Across all of this work, I take what I've called a Marian neurophenomenological approach, or Neil Seth calls it computational neurophenomenology, which I think is a better name.

where you can simultaneously analyze the system on computational or functional levels, what the system, what the different aspects are for, their adaptive significance, what their, the function they're serving.

And so the idea is you take this multi-stack understanding of cognitive systems, and you cross-reference this with phenomenology, taking experience seriously.

And active inference, with its associated process theories, basically checks boxes across all those levels.

You have the free energy principle, you have active inference, and you have predictive coding.

And these various claims of different degrees of specificity really help to flesh out this mutually constraining multi-level account of cognition.

But more recently, I have been collaborating with roboticists.

And here are two particularly excellent biobots, Tim Verbalen and Ozen Katal.

And along these journeys, we encountered each other, and there's increasing interest in machine learning and artificial intelligence and things like artificial consciousness.

You'll have people like Ben Gio talking about system two cognition.

And more recently, even Lagoon has been starting to take these things on with his JEPA architecture, talking about world models of different kinds and how they might serve the functions of artificial consciousness.

But for my collaboration with Tim and Ozen, we've mostly been focused on the specific problem of navigation, specifically a problem known as simultaneous localization and mapping.

Roughly the idea here is that where am I in space and what kind of space is this?

This is a fundamental question for any active inference system, any cybernetic system.

But there's a kind of a problem here in that you're trying to map out the space and situate yourself in it in the same time simultaneously.

So, but to map out the space, it helps to know where you are in the space.

But to know where you are in the space, it helps to have a good map of that space.

And so there's various proposals for different ways of bootstrapping yourself to degrees of certainty of what kind of maps or models are adequate for situating you in the world in a particular location.

And this is going to be important whether you're a robot navigating through the world or an animal foraging for value.

You got to know where you are to know what to do.

So, more recently, in this collaboration, which I've really found to be amazing because I just, I'm in awe of roboticists and what they can do.

It's kind of like that Feynman quote of what I cannot build, I cannot build, I do not understand.

Well, they live that.

That's just what they do all day long.

They're like introspecting, how do I work?

They're going into theory and then they're actually building a system in this back and forth iterative process.

And so the collaboration here was looking at what ways can biological systems and specifically focusing on the hippocampal system, what ways can these inform navigation problems?

In this more recent paper that I'm going to be focusing on more today called Generalized Simultaneous Localization and Mapping or GSLAM, we're reversing the flow a bit.

And the idea is more than cognitive neuroscience informing robotics, it's looking more to robotics to inform cognitive neuroscience, basically leveraging the precision they have with characterizing these problems and the way they're in touch with the ground truth of engineering and physical moving systems in the world, using this to inform cognitive neuroscience.

And specifically the claim is that this slam problem is fundamental and that it may have structured a

surprisingly, maybe shockingly broad range of cognitive processes.

And that this might be a fundamental, a source of fundamental principles for understanding how high level cognition was achieved with high level put in some scare quotes, but the ideas, the things that humans seem to be, or animals seem to be particularly capable of doing relative artificial systems and humans in some ways, even more so.

And the properties we want to recapitulate in artificial intelligence and machine learning.

So some precedent for this came from work with Ratslam, a seminal work by Milford Wyeth and Fraser.

So in this, they look to the rat and its hippocampus and look at its wayfinding and navigation abilities and try to basically reverse engineer these.

There's a really excellent YouTube video that I'll post.

I've watched it countless times, which just summarizes this process of building up Ratslam from like these initial foundations of the ambition and folding in more and more details, finding, you know, this isn't quite working, going back to the biology, going back to engineering.

And then actually showing systems that do the same things as biological systems inspired by those systems.

It's extremely compelling.

It kind of like, you know, every time I watch that video, I feel like I get something new.

I wonder, are these the steps that nature took in evolving these systems?

I also kind of felt like, what have I been doing with my life all this time?

Like, why didn't I find roboticists sooner and started talking to them?

So I'm very grateful to be part of this symposium.

Latent and very grateful for Tim and Ozen.

And so with their work with Latent SLAM, what they're doing is they're tackling the SLAM problem using active inference.

I'm not going to go into all the details here, but some of this was discussed.

There's parallels with some of the architectures that were discussed previously, and Tim touched on this.

So you basically have an agent moving through the world that has a variety of views of what it's expecting to see and poses.

And this experience map is a trajectory of the agent, room, space, and time.

And then here would be a generative model description in the middle of this, where you have these conjunctions of poses and views, which are unfolding in this series of transitions, adaptively sculpted via your policies.

But where there's this upper level, which is basically organizing this as spatial temporal trajectories, situating you in some kind of map or graph of space.

And this conjunction of pose and view, one of the places I found this to be compelling is that I've arrived at these as basically the primary modalities I focus on just from like looking at the systems neuroscience.

But this happens to be the primary things that are focused on for helping a robot to move through the world.

And pose and view, and basically for neuroscience, I focus a lot on systems such as the procuneus as potentially enabling something like a mind's eye and the lateral prior cortices as something like a lived body.

And that these would couple very closely.

And that would make some sense since your pose would inform your view and vice versa.

But there's another detail you need for this to be an autonomous system, an adaptive agent, which is not just like where's my body and what am I seeing, but where am I in space?

And where do I want to go relative to the world and the various things that I believe are in the world?

How do I pursue my goals in the world?

And so enter the SLAM problem.

But what's built with latent SLAM, this is a very impressive architecture.

Some details were described below, which all these things govern by a singular objective function where the agent is able to not just make its way through the world,

but also it's this powerful self-supervised learning curriculum as you're moving between views and pose and landmarks,

each one providing a source of predictions for the other, training up the other, helping you to get a grip on the world.

In terms of SLAM, two of the things that, so this like broader notion of G-SLAM or generalized SLAM, this would be a way of thinking of high-level cognition.

I'm particularly interested in these two operations.

One's called loop closure and the other's graph relaxation slash node duplication.

I guess that's three.

But the idea is, as Tim mentioned, when an agent finds itself in a situation where its pose and view are highly similar to something,

to what was previously encountered, this affords a potentially unique opportunity.

You now think you know where you are in space.

And so this is a really good opportunity to update your experience map.

And so this coming back around, it's not just a good opportunity in terms of you have high confidence, but it creates a closed system.

So you closed a loop.

And so you basically have these different equilibrium points in this graph, which can be thought of as a map also.

But basically, because they're all chained together,

these are sometimes modeled as like energetic spring systems.

And they're all mutually informing and constraining one another.

And so when you find yourself with a loop closure event where you're in this highly familiar state, you then take this as an opportunity to refine your map.

And so you will do a very fine.

If you're within a certain margin of error of a location,

if you think you really got it,

you'll then take the overall map and relax.

And you'll basically move around the various dependencies of different nodes and space, the different landmarks to adjust them and have each one refine the other and form the other.

However, if you're past some threshold for similarity,

and it looks like you're close, but not quite there, you're a little bit different.

If you're surprised, then you'll duplicate the node and then you'll create a new landmark.

And so now you'll have a new graph.

And so this is basically a kind of structural learning.

And there's a lot of work in Bayesian cognitive science on,

and within active inference, a lot of open work,

like how do we effectively do structural learning for handling complex domains?

And I'm suggesting, and I think Tim knows, hopefully would agree,

especially because we're writing together,

would be that this provides a principled basis for doing structural learning.

What's interesting here, though, in this diagram that Tim made,

is that as you're mapping out this warehouse here,

if you set these thresholds for identifying a node differently,

if your graph relaxation slash node duplication thresholds are set differently,

you get very different models or maps.

You might get very different structures, or if you will, schemas or scripts.

That's not the frame in terms of a SLAM problem,

but the claim here is that basically the hippocampal supposed place fields

provide a basis for structure representations,

many of which have resemblances to the kinds of things that were hoped for

in like good old-fashioned cognitive science.

And a neuro-symbolic AI.

And so some of, and part of what's hoped for is some of the aspects

of potentially uniquely human cognition, high-level cognition, high-level,

require some sort of representational scheme.

But if you set these parameters,

but if this is the mean by which you arrive at representations,

and if these parameters set differently could give you wildly different representations,

these would have very different impacts on the kinds of minds you might have,

potentially including even things like cognitive spectrums,

if you would apply this to a biological domain.

And so there's some notion that, for instance,

you could even potentially think of these parameters for the hippocampal system,

which are maybe evolutionarily,

were originally laid down as parameters for a SLAM problem,

for an animal finding itself in the world,

but where they continue to operate in terms of the mechanisms of hippocampal and entorhinal plasticity,

that these, if they're parameterized differently in different people,

could give you very different kinds of minds.

And this potentially has very,

the significance of this might be difficult to overstate
in terms of characterizing human difference.

These might be some of the most important variables
you could look at for saying how people differ
in the way they relate to knowledge in the world.

And not just, so this would have clinical consequences,
basic science consequences,
maybe even helping to explain some of uniquely human cognition,
the kinds of things that help to get symbolic reasoning
and like cumulative culture off the ground.

There's other proposals that are kind of similar to this
mad dog navigation perspective or SLAM perspective.

Jeff Hawkins out of Numenta has a proposal he calls a thousand brains theory.

Technically it should be called something more like a hundred thousand brains theory,
where the idea is that there's a common cortical algorithm
carried out by every single cortical column
of doing this object modeling and navigation
of the kind described with the pecanple and entorhinal system.

I find aspects of this to be compelling,
but also very mixed feelings.

And I can talk about that later with anyone who's interested.

There's another recent proposal in this spirit
by Chris Fields and Michael Levin,
where they're basically characterizing morphogenesis
as cells engaging in a kind of wayfinding
and navigation process of where they ought to be
in this synchronization manifold of the phenotype.

And so they're all finding their place,
you know, singing as Carl might say,
like singing from the common hymn sheet,
knowing where they are.

But as they're synchronizing,
they're moving around to find where they are.

This unfolding geometric melody
that eventually becomes the phenotype of the organism.

But the idea is that this intelligence of wayfinding at the cellular level,

many of these mechanisms may have been repurposed
and recycled and deployed differently for overt navigation.
And that this is a frame for intelligence more generally.
I find this to be extremely compelling.
Although what I'm doing with this work
is I'm focusing more on the hippocampal and entorhinal system
as a source of specific adaptations with specific properties
and a major transition in evolution.
Potentially also with some connection to things like
Ginsburg and Jablanka's major transition markers
and unlimited associative learning.
They have a lot of emphasis on the hippocampal system and its homologs.
There's very interesting work that I'm only beginning to look into
in the insect, in insects with the central complex
and the mushroom bodies and the way they handle navigation.
I think cross-referencing the mammalian version
and the insect version would be really invaluable.
And that's like, that's a major to-do
that I'm hoping Daniel will help me with.
So some other precedents would be in terms of this idea
of generalized navigation,
that this would be a powerful frame for understanding cognition.
It comes from some older work with Hassabis and McGuire.
They found interestingly that hippocampal patients,
not only are they impaired in episodic memory,
but they have trouble with episodic imaginings
of counterfactuals of novel situations.
And so they talked about this construction system of this brain,
this core system for memory and imagining,
allowing for various forms of sophisticated modeling
and creative cognition.
And this seems to have been part of what basically gave Demis
the confidence to go and found DeepMind.
And this is still a core aspect of what they're doing there.
And the race is on for reverse engineering the system
and the race is heating up.
And some really excellent work would come from Timothy Barron's group.
I believe he's still at UCL.

They're showing like, for instance, there's an architecture called the Tolman-Eichenbaum machine, which I strongly recommend checking out.

That's active inference compliant.

And as a realization of the function of the hippocampal system, they don't focus as much on navigation, although it can be used for that.

And that's part of it.

I'll come back in a second for the differences here.

But in this work, one thing of note is so they're showing basically people categorizing these stimuli where you have these birds with different length necks.

And there's a kind of morpho space in terms of characterizing the features and their various dimensions.

And basically they're showing hippocampal slash entorhinal-like representations, grid-like representations, being involved in helping to map out this feature space.

And so this is one example that's very commonly cited of spatialized cognition of something that might not seem clearly spatial.

But then the idea of G-Slam would be once we've spatialized it, then to what degree in our sense making, is it actually kind of navigation through the space?

And so this process of spatializing and navigating through the space simultaneously, I would call that a SLAM problem.

And I think a lot of cognition can be fruitfully framed in this way.

And I'm going to go a little further.

So here's the Tolman Eichenbaum machine and he's showing like different,

I don't know, is his face covered by the thing?

I can't tell.

Well, there he is.

Yeah.

So his work is unbelievable, his group.

But different things like relations among people within hierarchical tree structures of their relations, doing things like transitive inference of one property related to another and a set of dependencies, but basically using the hippocampal systems properties to navigate and manipulate these spatialized representations.

There's other work from DeepMind.

DeepMind. There's this other talk highly recommended by Stachenfeld et al. at the Maine conference. It's unbelievable talk.

Talking about the recent work from DeepMind to reverse engineer the hippocampal systems, place cells and the grid cells of the entorhinal system, as allowing for a variety of properties.

Some of, you know, very similar to Barron's work and they collaborate together.

But one thing that's interesting about this talk is she makes a point that this isn't just for tasks that are like physical space.

This isn't necessarily spatial.

I might challenge that a bit and say that we should basically always have a prior of looking for the spatial properties of any task domain and then wondering to what degree can we frame this as a SLAM problem. And I think there's maybe two reasons of doing for doing this. One, maybe three. One is that basically putting things into a common framework helps to you to draw analogies across seemingly heterogeneous domains. And this can be a source of insight in terms of a common representational framework for sense making. Sense making gives you an ability to have creative combinations from seemingly different domains. But the other reason is the hippocampal system, like this was specifically probably what it was selected for. This is a mapping system for a animal forging through value through various spaces. And this isn't just evolutionary, but this is also developmentally and ongoing. This is an ongoing problem. We have not transcended this. We are still in space. We still constantly have to move through space and situate ourselves through spaces, physical spaces, and also conceptual spaces. So that's part of the reason I'm really emphasizing the SLAM framing because the hippocampal system will never lose this job description. And this, the idea is that understanding the ways in which it's parameterized, specific operas, operationalizations, I don't think that's a word. It's essential to keep in mind this ecological significance. And so this is all within the spirit of, I guess, a broader view on cognitive science called ecological rationality, which is go back to the origins and try to understand it's Tim Bergen's quote, nothing in biology makes sense except of the light of evolution. Take that very seriously and run with it. One other thing that's, uh, so alongside one thing I think you get is connections. One thing a G SLAM perspective gets you is connections to foraging theory. And so, um, you know, if you're thinking of an animal going through the world, trying to extract value, um, resources are often patchy. And so as you're moving through the world, you, uh, might be, you know, balancing exploration and exploitation with respect. If you find a patch might be good to exploit that for a bit if it's particularly juicy, but then as that patch dries up, maybe it's time to go explore. And then you want to move between adaptively move between phases of exploration, exploitation, staying longer in some places and then moving to other places. And so these, um, jumps other places, like when a patch starts to dry up, and these are sometimes known as levee flights. And, uh, this is playing actually into some of my work in psychedelics and that, um, informed by some really intriguing suggestions from Matt Shine's group. Uh, the idea is that's one of the mechanisms impacted by psychedelics might be, um, potentially facilitating these kinds of levee flights, where you can think of more creative cognition as involving more exploration, more broad associations. And there's all, and foraging theory, uh, you know, it's, it's extremely rich and well-developed. Uh, there's all sorts of different modes of foraging that could be thought of as different modes of cognition. There's specific hypotheses and connections to formalisms like, um, the marginal value theorem. So you would have like a hypothesis of like, you know, when do you leave?

Well, in the marginal value theorem, that is you leave a patch or when do you, when do you go and try to find a new, you leave a patch when the average, when your current estimated value drops below what you think the average rate of return should be. So this then, you know, gives you, um, basically like you can now characterize people, where are they relative to these different normative models? Um, and you know, this idea maybe has an even older vintage or, or maybe, you know, newer relative evolution, but old relative to discourses in terms of the art of memory and the art of memory and the method of loci, you know, memory, especially before we wrote everything down or when writing things down was expensive. And even now to have something in your mind where, you know, it could be at hand where you can retrieve it to have like a well curated set of, of knowledge, it's essential. And so one of the best tricks that were discovered is you actively intentionally spatialize your knowledge. You actually create a physical place that you visualize and you move through it. And at each place you've, at each point in this space, you've placed different things you want to remember. And part of the reason this likely works so well, like almost certainly is this is, um, you know, the core of memory in the hippocampus system is actually involving a process of, um, it is a spatial mapper. That's something I would argue. Uh, there's connection to some other, um, work cognitive science, I'm not gonna get into right now, but like multi-dimensional scaling, where you understand potentially complex domains by mapping them onto a more, uh, a simpler and potentially more, uh, cognitively navigable space. Um, and then I'm not gonna get into this right now, but, um, actually I will. So, uh, real quick aside is with respect to, um, uh, multi-dimensional scaling, one of the early things that got people excited for it, about it was it's potentially used for analogical reasoning or establishing similarities between domains. It's, it's a very important thing to know, like how similar is something relative to something else? Like, should these things be placed in the same category or, um, you know, it's, and one way of doing this was saying, what is the distance in this, when you do multi-dimensional scaling and you place something in a feature space, what is the distance between these, between different things? And the further apart, the less similar they are, but there are other models of analogical reasoning and there's another, um, race on to try to figure out how we do analogical reasoning. And so some of these proposals, um, you know, they would go on beyond something like, um, multi-dimensional scaling, but they involve specific structure representations. But then we have a question, well, what's the ontological status of such things? Do they actually exist in the minds? We do actually have representations, or maybe these things are more like they're virtualized or inactively realized or implicitly. I suspect they probably, we probably, um, many of the kinds of structures, you know, described in good old fashioned cognitive science might not be biologically plausible, but they might be surprisingly biological plausible. If you actually have a system capable of adaptively constructing graphs based on, um, your ability to extract value from these graphs and situate things relative to other things. And so it's possible that some of analogical reasoning and even causal reasoning could be unlocked by having a good means of, um, creating structured representations. And one of the things that the Campbell system might

seems to be able to do is create these graphs where the different place cells of the of the hippocampus can be thought of as nodes. And so if you duplicate a node, you are now basically doing structural learning. Um, you can, you have, you're complexifying this, uh, representation. Uh, so this comes into my, uh, work in other ways, such as, um, models of goal-oriented behavior. Uh, I'm not gonna get into too many details right now, but here would be the campus system. And you see, um, based on work of people like reddish that, and also, um, like Sam Gershman has a successor representation model, but basically you'll see these predictive, that campus is a predictive map, where what's being predicted is, um, where you'll see these sweeps of activity that will proceed in front of the animal before, and then we'll predict what choice it will do, what arm of a maze it might go to. And so the idea is that the hippocampal system is the top of the cortical hierarchy and that any prediction errors that are not handled elsewhere end up get temporarily stored in this volatile, uh, memory system, but which is also encoding these trajectories or these series of equilibrium points of state transitions. And that, um, basically you have, you know, prediction errors that are not handled elsewhere, make their way there, get temporarily stored. And you can have, um, what's sometimes called semantic pointer architecture, where basically as you're moving, as you're stepping through these various, um, equilibrium points, you can then unpack this as estimates of system and world, um, in terms of operative policies and actions and what you expect at different points along these different trajectories. And so here I'm focusing on moving through a physical space in, um, this case making tea, but this is, I would argue, this is also part of what we do when we navigate conceptual spaces that we are, um, deciding what branches do we go down to what degree, where will we find what kinds of value at that? This is a similar operation using similar, uh, procedures from common neural processes. Uh, more recently, I've been trying to extend this to, um, models of, uh, volition, including, um, free will. And let's see.

I don't, so, uh, Daniel, how much time do I have left? I'm sorry.

Ten minutes? Okay, great.

So,
this is, uh, less, um, I guess explicitly, um, slam framed, um, but still it's a, um, hippocampus at the center of it. And so, so here, um, I'm trying to, this is a recent piece, it's in preprint form, which I'd have, uh, I'd love feedback on, but basically the idea is as your, that the present moment of experience, uh, has a thickness to it. Uh, this is sometimes called the specious present or the remembered present. And that, uh, it seems to vary that the minimal thickness of a moment of sense-making seems to be about a third of a second. And the upper bound seems to be about three seconds. And that this seems to correspond to, um, an epic. So, so you, you basically can, so the hippocampus system can, whenever you enter like a room, it'll tile that domain with a certain degree of granularity, it'll create these hexagonal tilings of that space. And then within there, that's where you'll have these different attracting attractors, these different trajectories play out, orchestrating the rest of the brain in terms of its dynamics to pursue these various goals. And so the runway you have to work with in terms of the thickness of the different

rollouts for sophisticated inference would depend on, um, partially, uh, the shelf life of this set of attractors before you have to refresh them, uh, in some way, either set down an entirely new set or, um, uh, try to recreate the old one or prevent the remapping. And this would be related to some of what was just described with air dynamics in terms of how good of a grip do I have? So let's say you have like a poor grip on what's happening and your prediction error is going up. This is not a good situation for you. That might be a good opportunity to stop, hold, and catch fire, retile space, get a new set of operative policies for sophisticated inference. And so, um, so here what I'm trying to do is show that um, different levels of, uh, agonism of these serotonergic receptors involved with psychedelics, 5-HT_{2A} receptors are influencing the stability of these sets of attractors and the vividness with which you are doing these imaginings and of rollouts of actions into the future. And here it's, you know, it seems that what you could potentially do is describe some of a dose response curve for something like psychedelics, where as the dose increases, the extent of the rollouts become greater. You've stabilized them and imagination become, can become more vivid and can influence your

policy selection more because you have more confidence in what you're seeing. And so sophisticated inference, your imaginings are contributing more to your ongoing overt action selection up to a point. But then, and so this would be, and so this, but then at a certain point, um, the idea is you keep dosing, uh, the animal or maybe the robot and it enters more of a creative dreamlike regime where it starts to lose coherence and you see the, of the extent of these rollouts of policy selection decreasing, uh, things are becoming more vivid, but less coherent. And sometimes you're getting, uh, and this itself could occasion error and potentially re-gripping in this model. And so the idea though is that this would be a cognitive spectrum potentially, in addition to explaining some of the neuro-phenomenology of things like psychedelics, this could potentially explain some cognitive spectrums, whether we're talking about um, maturation or aging or waking up, you know, when you first wake up to going to sleep, these different parameterizations of your mechanisms for sophisticated inference, helping to basically provide a neuro-phenomenological handling of human, uh, of what it's like to be a person and all the variations of it. And so I think that should be it for right now. And, uh, happy to talk about all this with anyone who's interested. Thank you. Awesome. Thank you, Adam.

Hi, Daniel. Hey, any, you're muted. Thank you very much. Any other, uh, notes you'd like to add?

Um, I think that's, uh, I think that's it for the moment. Great. I hope that you, uh, consider anything and join us for the round table shortly. So thanks. And we'll be right back with the final presentation by GF Clatcher.

Thank you.

Right.

All right, we are back.

This will be the final presentation of this interval.

This is JF Cloucher towards a symbolic implementation of active inference for LEGO robots.

So looking forward to hearing this and then seeing you on the round table JF.

I'm Jean-François and I will talk about how I program LEGO robots as active inference agents.

This will contrast with most robotic implementations of active inference which are based on probabilistic inference.

So I hope this will spark some interesting discussions.

Well, first of all, I'm a software developer.

I'm not a neuroscientist.

In my spare time, well, I like to code models of cognition because I'm keenly interested in all matters of cognition and consciousness.

And I run them on robots because the real world is hard.

And I believe that writing code for robots is actually a very good way to test my understanding.

And coding itself helps make abstract concepts tangible to me.

So what do I want to accomplish?

Well, I want my robots to learn autonomously while they conform to the principles of active inference.

This is the current version of my robots.

There are two earlier versions and there's one in early development.

And here we'll see my robots to the left, Carl to the right, ND, driven by the interplay of generative models for hunger, danger avoidance, and wanderlust.

Let's see them in action.

They navigate a space.

The sheet of paper on the floor represents food and they are attracted by the scent of food simulated by an infrared beacon on a pedestal.

So Carl is making a beeline to the food while ND is having a problem getting traction and bumping into the walls.

Carl is cautiously approaching, but we'll get a little bit too close to the pedestal.

And this will engage its danger avoidance, collision avoidance, generative models.

He backs out.

Andy is observing all this.

Andy backs out as well.

Carl is re-approaching the food.

I'm hungry.

And this time we'll eat successfully.

Carl is freaking out.

I'm hungry.

And he has noticed that Carl was freaking out.

And in sympathy, he freaks out himself while Carl is eating.

Alright.

So that was the robots as they are right now after three iterations of this project.

So what does it take to implement an active inference agent?

Well, there are three approaches.

One, let me take the numerical approach, which basically puts the mathematics of active inference agent into code and runs that on robots.

Another approach is to take a symbolic approach where we have code functions that operate mostly on symbols, predicates, etc.

Here we can see we have the symbols for hungry, very, danger, curiosity, etc.

But that's not the only option.

Take E. coli.

E. coli is an active inference agent and its implementation is biochemical.

Well, I clearly didn't take that route.

I went the symbolic route.

Well, mostly because it's the familiar one to me and it provides me with a better scaffolding for my intuitions and exploration.

I also find that it handles complexity quite well.

When I started this project in 2017, my first thoughts were to implement a society of mind.

I'd come across this concept decades ago.

It's a theory of mind put forward by Marvin Minsky 50 years ago.

And at its core, it says that intelligent behavior emerges from a lot of simple actors that interact in simple ways.

It so happens that I look at it as an architectural principle as well.

So how does one implement a society of mind?

Well, to me, the answer is clear is we use the actor model.

The actor model is a model of concurrent computation that was introduced by Carl Hewitt in 1973.

According to this model of computation, we have independent actors, which are processes, concurrent processes,

each one managing its own state.

And the only way an actor can interact or influence another actor is by sending it messages.

And once an actor receives a message, it will interpret that message, possibly modify its internal state, and as a consequence, fire messages to other actors.

So building a society of mind using the actor model is quite a natural fit.

I also happen to program in the Elixir language.

And Elixir is a message-oriented multiprocess language, which is based on the actor model.

So it's a perfect fit.

So here we are.

It's 2017.

I own two Lego Mindstorms EB3 kits.

They're wonderful robotics kits with a nice choice of sensors, mini motors, and a computer, the EB3 brick that controls them all.

And I want to build a society of mind in Elixir on my EB3 robots.

I'm in luck.

There's a Linux distribution called EB3 Dev that allows me to run Elixir on the EB3 brick.

So I'm all set.

So that allowed me to start this project.

And this project, as I said, has completed three iterations, and I'm starting a fourth one.

And we look at each one in turn, reviewing the goals and also the issues uncovered.

First iteration, 2017, version one.

A simple society of mind, as I call it.

So I implement an ad hoc model of cognition.

It's not something that I took from research.

It's something I made up myself.

It's homebrewed.

And let's have a quick tour.

So being a society of mind, it's populated with actors.

And we have the detector actors, and their responsibilities are to interface with the sensors and to pull them periodically to generate percepts, which are events that are, like all other events that the actors will produce, are sent to the

the central nervous system actor, which is a dispatcher, which will then dispatch these events to other actors that are interested in them.

So there's a memory actor.

There's a memory actor, and all events are sent to the memory as well, to be stored for a certain amount of time, and represent the past, if you want, to the agent.

And the other actors are able to query this memory in order to make whatever decisions they need to make.

So one of the key actors are the perceptor actors.

And what they do is they ingest events, percept events, originally from the detectors, and analyze them, maybe in the context of the past, and produce higher level percepts.

For example, a perceptor might be digesting a distance percept from a detector, look at it and say, ah, given prior distances, I'm getting closer to an obstacle, and then produce that percept, and the higher level perceptor might say, ah, I am getting closer to an obstacle, and I am currently close to it.

So, ah, produce a percept that says collision imminent, and so forth, and so on.

Another important kind of actor is the motivator, and the motivator is responsible for deciding what the robot needs, what the robot wants, and it expresses this need or this want as a motive event, that is, again, sent to the central nervous system, dispatched, in this case, to the behavior actor, and the behavior actor then takes this motive and

acts on it by emitting intent events that are listened to by the actuator actors who translate these intents into actual commands and move the robot.

One key point is that the motivators actually compete with one another so that the most important need of the robot is the one that's expressed.

For example, getting to safety is more important than satisfying hunger.

Let's look a little bit into one of the types of actors, the behavior actors, and the behavior actor is actually quite complex.

It does a lot of work. It's not a simple actor. It's implemented as a state machine, and with every state transition triggered by new percepts, intents are emitted.

And so we can have a behavior actor, which job is to direct the robot to the food.

So, it works. So let's see ND version one, moving about and trying to get the food, which in this case is the blue paper on the ground.

So, it tries to avoid collisions. It does not always succeed. Here it finds out it's stuck.

Backs out. Goes around. Doing a better job at avoiding obstacles. And now makes a beeline to the food.

Having detected it. Approaches cautiously and feeds.

All right. So, this worked quite well, but there were issues with this version one.

Not, this is, I don't think it was visible in the video, but the robot was frequently overwhelmed by a constant flow of percepts.

The perceptor, the detectors constantly feed percepts into the society of mind.

And after a while, they kind of back up and the robot starts reacting to old events.

And one solution was to actually put the robot to sleep once in a while, paralyze it, let the events wash through the system, and then wake it up with a clean slate.

So, that was kind of a little bit awkward. And another issue was that, well, there's no learning going on.

The robot doesn't get better with each run. And given the model, it wasn't clear to me where I could fit learning in there.

So, time for version two. So, I need to find a way to make my robots learn.

And they also need to better focus and be able of attention so they can avoid the overwhelm with all these unneeded sensations.

I also need more computing power, so I replaced the EV3 brick by a Raspberry Pi 3.

About the same time, I come across Andy Clark's book, Surfing Uncertainty.

And I find that the predictive brain makes quite a lot of sense to me.

I think it tells a really good story about learning and attention, so I decide to implement a predictive society of mind.

This is my first crack at it. I won't go into the details.

I do in a companion paper.

Let's just say that there are actors for managing beliefs, validating them, making predictions, and taking action to make these predictions come true.

There's also an attention actor for turning detectors on and off as needed, and a focus actor for prioritizing belief validation.

I do introduce some learning. The robot learns what actions correlate well with success.

But that's about it.

I've got it.

And there's... the model does not feel quite right to me at this point.

And they are bottleneck actors.

For example, there's only one experienced actor.

There's only one attention actor.

Only one memory, of course, and only one focus.

And in his side of mind, that just doesn't feel right.

Yep, it still... it works.

And let's watch ND on its first run.

It's ND version 2. Hasn't learned anything yet.

So, same scenario.

So, ND has detected the food.

Would like to move towards it, but doesn't know which actions validate, believe, best.

It smells food, but still, you know, it's just heading into the wall.

So, poor ND is being quite unproductive here in its quest for food.

It's a little bit painful to watch, but stop.

Now, he's ND 30 runs later.

And it's much more competent about matching beliefs and actions.

Nels food and very competently makes a beeline to that food.

Just...

Here we go.

Obviously a very different ND from the naïve one.

And there we go.

there we go. Okay, so it worked, but there's still, there were issues with version 2. First, the model doesn't quite feel right to me. It doesn't quite capture something I didn't feel right about it. And there were bottleneck actors as well, which is not a good sign. And learning was limited to action selection. So time for version 3. I call this one a society of generative models. So I go back to Andy Clark's book and also read on the free energy principle. And it becomes clear to me that generative models are actually quite central. In fact, it's generative models all the way down. So I get to work on a new model.

In this new model, I have very few kinds of actors. There are detector actors, as before, to interface with the sensors. Actuator actors to interface with motor speakers and whatnot. And the only other kind of actor are the generative model actors, the GM actors.

And there's quite a few of them. In this case here, 16. And they do the heavy lifting. They do the belief updating, the predicting, the raising of prediction errors, the selection of actions. And they are responsible for remembering their past states and actions. Each GM operates within its assigned scope. There's one for locating food, one for avoiding obstacles, one for detecting approach

obstacles, and so forth and so on. The GM actors have parent-child relationships.

A parent-GM sends predictions as events about what it expects its children, GMs, to believe, even what it itself believes. And these prediction errors flow downward. And prediction errors flow upward.

If a child GM receives a prediction about what it's supposed to believe, or it's expected to believe, and it's not what it believes, and it sends back a prediction error up to its parents.

Well, that means that a generative model's perceptions are, in fact, the uncontested predictions it makes, plus the prediction errors it receives. And the GM updates its beliefs based on these perceptions.

And given what it believes, it may decide to take some actions and then raise intent events that reach the actuators. A little bit like before. The generative models are kind of grouped into different areas of concern.

There was, there's being, which is responsible for deciding whether we're in danger, we're hungry, or we're free.

There are the danger GMs, which have to do with clearing obstacles, trying to stay in well-lit areas, avoiding collisions with the other robot, and whatnot. Then there are hunger GMs, that have to do with eating, seeking food, etc.

There's the freedom GM, which I call the wonderlust GM, which basically says, well, I'm neither in danger nor hungry, so let me just roam around.

And finally, there are these two ones here, which have to do with guessing the intent of the other robot.

And we'll go into more details with these two.

So, let's look more closely at the GM actor.

Remember that a GM has a, as parent GMs and a child GMs, that predictions flow downward and prediction errors flow upwards.

Let's look inside.

A GM defines one or more conjectures.

A conjecture is a potential belief.

For example, whether I know where food is or whether I'm about to collide.

Two different conjectural beliefs.

A belief may either be a goal, that's a belief we want to make true, like I'm eating, or an opinion, which is a belief that I want to validate, for example, that I am approaching food.

A conjecture is typically activated when the GM receives a prediction about the conjectured belief.

And a conjecture knows how to check its associated belief to see if it's valid.

It knows how to make predictions given that belief,

and how to decide what actions to take to either achieve or test the belief.

A GM learns how well alternate courses of actions, policies, correlate to achieving, validating the belief.

It also keeps track of which child GM it trusts more,

should they send competing prediction errors, so it does precision weighing.

And each GM actor works in its own time.

It's a concurrent process.

And it does one, it does, it operates one round at a time.

And we'll see what a round means.

And it remembers its past rounds.

Its past perceptions, the predictions that's perceived in the past, its past beliefs, and actions.

So now let's look at the life cycle of a GM actor.

Well, as I said, a GM actor operates round after round after round.

So what's a round?

First, when a round is started, a GM carries over the perceptions and beliefs from the previous round.

It doesn't start from scratch.

And if it's in the midst of trying to achieve a goal belief, it may, as it starts its new round, activate conjectures for that action belief.

And as a result, send predictions to child GMs.

Once we're done starting, the round is active.

And then the GM actor just sits there for maybe, I don't know, 300 milliseconds waiting for events.

Events like predictions coming from above, from parents.

That will then activate conjectures trying to prove that the predictive belief is actually true.

This will cause predictions to be sent to child GMs.

And as a consequence, maybe prediction errors will come up.

And the GM will update its perceptions.

When the round is completed, either because we've heard from all the children or because time's up, the GM completes this current round.

It updates position weighing, keeps its trusted perceptions, updates its beliefs.

As a result, may send prediction errors because new beliefs conflict with predictions that it had received before.

It updates its policies.

It decides on the course of action.

It sends intents and notifies the round is completed.

And we continue.

So round after round after round.

I mentioned that there were two very kind of more interesting GMs in the Society of Generative Models.

These are these two GMs.

They represent my little foray into theory of mind.

These GMs are dedicated to guessing what the other robot is doing.

So each robot observes the other and detects patterns of movements and infer intentions from these patterns.

There's only two kinds of intents that are inferred.

One is that the other robot has detected food and is making a move towards food.

Or that the other robot is in danger and is panicking.

Now, how does a robot see the other robot?

Well, in addition to the array of sensors I already had from previous version, I've added a new sensor, a 360 beacon seeker.

And what this beacon seeker sees is the beacon on the back of the other robot.

So each robot has a beacon on its back that broadcasts in a 360 radius circumference angle.

It signifies, here I am, here I am, here I am.

And the other robot can detect how far the other robot is and at which relative orientation.

So let's say one robot here, Carl, is making a beeline to the food.

Andy has been observing this and seeing the pattern of movement, which is a very straight trajectory.

Infers that Carl has detected food and is homing in on it and may decide to follow Carl and get to the food as well.

Another pattern of movements would be a chaotic pattern of movement.

And let's say Carl had been backing away in panic from hitting the pedestal.

And Andy was observing this and noticed that this kind of chaotic pattern and inferred that Carl was in a panic and decided to get in a panic as well, if you remember the video at the beginning of the presentation.

So this is going well, but we all know debugging is hard.

I find out also that debugging societies of mind on robots is very hard.

In order to help myself, I think of it as simulated embodiment, which allows me to observe what goes on inside a given robot.

It helps for debugging, it helps for experimentation.

In the simulation world, my robots, virtual robots live in a grid world.

There are obstacles, the blue tiles.

There are kind of darker areas that the robot tries to avoid, which are represented in the black and gray squares.

And then there's the food, which is the green tiles here.

As the robots navigate that space by using their society of generative models, I can see where each robot is in terms of its state.

First, I can see where it's located.

I can see its orientation, what it's last said, what the last action it executed, and I can see the state of its virtual sensors and virtual actuators.

I can also peek inside the various generative models of a robot.

Here I'm looking into the avoiding obstacle GM of Andy.

And I can look at the state in the current round or in past rounds, and I can see which predictions came in, which conjectures are activated, what are the current perceptions, what beliefs that the robot holds at the time, and what actions it thinks are more likely to work, and what actions it is taking.

Very, very helpful.

But this moves really fast, and so I can actually slow down the simulation and even pause it and then look inside the various generative models and see if what's happening is what I expected would happen.

So great debugging tool and experimentation tool.

So I'm quite happy with version 3.

I think it aligns quite well with the active inference ontology.

There are no bottleneck actors.

But learning is still limited to selecting action policies.

However, I think that version 3 sets the stage very well for the new iteration, and this one has an ambitious learning agenda, and it's just getting started.

So version 4, starting now, I call it active inferencing.

So here's what I want to achieve.

As the robot interacts with its environment and learns how to maintain it, I want for it to learn how to maintain its homeostasis by growing its own society of generative models.

So instead of being given an impriority set of generative models, I want the robot to learn that set.

Not only that, but I want each GM to learn its own capabilities.

I want it to infer its logic programs that, when executed, will infer predictions from perceptions, will infer beliefs also from its perceptions,

and infer policies that will validate or invalidate its belief.

So how does one do that?

Well, there's a very, very interesting paper that I found, which is called Making Sense of Sensory Input, written by Richard Evans et al.

from Deep Mind.

And it answers the question of how do you learn a causal theory that explains a sequence of sensations.

And once you've built that causal theory, you can then apply it to predict what the next round of sensations or perceptions will be,

and thus make predictions about them.

Just to read from the beginning of the paper, Making Sense involves constructing a symbolic causal theory that both explains the sensory sequence and also satisfies a set of unity conditions.

Now, unity conditions are actually a set of constraints derived from Kant's philosophy that ensure that all the pieces of the theory form a coherent whole, semantically speaking.

I won't go into the details, but it's quite fascinating.

The paper goes on.

Making sense of sensory input is a type of program synthesis, but it is unsupervised program synthesis.

So the paper presents an apperception engine.

This is the software that will infer causal theories from sensory data.

It is a code generator.

What it does is it synthesizes logic programs that implement these causal theories.

And when you run these logic programs, you can predict incoming sensations.

So essentially, an apperception engine generates predictors.

And the way it does that is it searches for predictor logic programs in a space of potential possible logic programs.

It's a search problem.

And the search space is extremely large, of course.

So we need a way to constrain it.

We need to apply strong inductive biases to restrict the search.

That's where the unity conditions come in.

So in order to derive its own predictor, a GM will search for good predictors using apperception.

So given a prior round of perceptions, remember that the GM has a memory of each round,

and in each round a set of simultaneous perceptions, given that sequence,

a GM will search a space of candidate predictors for a good one.

What makes a predictor?

Well, two things.

First, there's a scope, and then there's rules.

The scope is essentially the objects the rules will be about,

and the belief predicates that will be used in the rules.

The rules, well, there's three kinds of rules.

There's the static rules, which apply on a set of simultaneous perceptions, like at time n .

And the rules validate and possibly also infer implied perceptions, missing perceptions.

An example of a static rule might be if the robot is at distance zero from an obstacle,

it is also touching the obstacle.

Then there are causal rules, and the causal rules essentially infer the perceptions at time n plus one, given the perceptions at time n .

They predict the next perceptions.

An example of a causal rule would be if the robot is at distance x from an obstacle,

and the robot is approaching the obstacle.

In the next round of perception, the distance of the robot to the obstacle should be smaller.

And finally, there are conceptual rules, which are rules that kind of make the whole predictor hang together semantically.

There are rules of mutual exclusion.

For example, there can only be one distance between a robot and an obstacle.

And rules of uniqueness, which, I'm sorry, rule of uniqueness would be that they can only be one distance.

Yes.

Rule of uniqueness, there can only be one distance, for example, between two objects.

And mutual exclusion might be a robot cannot both approach an obstacle and be touching it at the same time.

So the apperception engine will kick in for a GM when the current predictor of the GM is no longer good enough for the job.

Like we've produced a predictor, and now we have more rounds of perceptions,

and the predictor is doing a really bad job at predicting the next perceptions.

And then the apperception engine will kick in and try to produce a better predictor for that GM.

So that happens for all the GMs.

And once we have a good predictor, the GM, even the current set of perceptions, can infer the next set of perceptions, thus make predictions.

So we talked about how a GM learns.

But what about how it learns, how about learning the entire society of GMs?

How does a robot form its society of GMs through interactions with the world?

Well, this was going to be guided by a mid-tech cognition GM.

That's something that's given when the robot starts life.

It has a basic set of detectors, set of actuators.

That is a given.

And it also has a metacognition GM.

It's also a given.

The metacognition GM, its role is to watch over and guide the evolution of a society of GMs.

That's the action.

For it to function, each GM must act as a cognitive sensor.

It must produce sensations such as, are prediction errors trending up or down?

Do policies predictably impact my beliefs?

Or, very importantly, within my purview, is the robot's homeostasis at risk?

These are cognitive sensations that a GM will produce, will be listened to by the mid-tech cognition GM.

And that will allow it to update its own beliefs about how well the society of GMs is doing.

Does it have sufficient coverage?

Is each GM capable enough?

And based on that, may take corrective actions by creating new GMs.

And possibly, if one GM is not doing very well, it's stuck in a rut, removing it and replacing it with another.

So that's the role of the mid-tech cognition GM.

So, how can the robot tell whether it's doing better or worse than before, given what it has learned?

Well, feelings.

That's how.

So, a robot will have internal sensors for feelings.

And I'm thinking about feelings of hunger, which increases with physical and mental exertion and is reduced with eating.

There's the feeling of pain, which would go up with a number of collisions, but would be reduced automatically with the passage of time, simulating healing.

And then I'm thinking also of the feeling of ennui, which goes up if a GM is not learning anything, if its logic programs are unchanged for a long period of time.

But ennui goes away when the GM is learning.

Feelings have a balance, positive, negative.

I'm hungry, very hungry, negative balance.

I'm not hungry at all, positive balance.

And a negative balance signals risks to homeostasis.

And when beliefs are derived from feelings with a negative balance or a positive balance, then these beliefs take on this balance.

And this is going to be used to prioritize actions because actions that promote beliefs with a strong balance will be prioritized for actions that do not.

So, in other words, a GM with very strong feelings will have its actions prioritized over GMs that currently want to take action, but from less powerful feelings.

And if these actions conflict, then the actions of the GM with stronger feelings will take over.

Well, given all this, what do I think makes a robot an active inference robot?

Well, there's many things, but I'd like to think that in its simplest expression, for me as well, at least, an active robot is one that learns what to do in order to feel good more often than not.

So, I'm hoping that this work will raise some big questions and give me some insights into them.

For example, what is the apparent knowledge that an agent must possess to learn autonomously?

What makes unsupervised learning converge on competency?

Diverge.

And maybe a little bit more far out, what can growing a society of GMs learn from developmental biology?

I suspect that there are principles at work here that they both share.

So that's it. Thank you very much.

Thank you to the Active Inference Institute for their help and encouragement.

The version 3 of my work is available on GitHub.

It's open source. I can be reached on Discord and by email.

And the companion paper to this talk is available on Zenodo.

All right. Great talk, JF.

We'll take a quick break and we'll be right back with the roundtable.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

And welcome to JF.

Carl will be joining shortly.

So perhaps to begin, if anybody would like to provide a reflection on a talk that was not their own.

What was something that you felt like was brought to the table?

What was a key problem area that was addressed and how did you see active inference being implemented and adding meaning in that situation?

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

I can maybe quickly just comment on JF's presentation since it was the most recent.

I think definitely this symbolic approach to active inference is one that is perhaps not been that pushed forward.

And like, obviously, I think, and like, obviously there are different approaches all around the field, but I feel like that this particular perspective on pursuing this kind of emergent generative models, which only have the very basic kind of setup, but then the complexity emerges through the usage of Elixir and Prologue, I think was really interesting.

And yeah, I mean, we talked about JS project in the organizational meeting units quite a lot.

But every time I think about it, it's really cool to see this implementation.

Thanks, Jacob. Adam?

Well, I mean, in terms of the recent one, just the app perception engine, like the Kantian inspiration that I found that to be really exciting.

Especially because, you know, I've been taking some inspiration from Kant, some of the work and like, wondering whether basically, like, these categories, like, are they not just like preconditions for judgment, but like any kind of sense making whatsoever, like any sort of world with composition that could be navigated and parsed.

And so, to see that that kind of work was being done with that level of sophistication was just amazing.

It is a thread that came up multiple times with juxtaposing discrete time, continuous time, and all of these.

And then it's almost like another dialectic with the numerical and the distributional approach to variational inference, sampling, message passing, a whole taxonomy of approaches that we've seen for active inference models.

And also models of perception, cognition, and control outside of active inference.

And so, maybe, JF, if you wanted just to provide a thought on where you see the symbolic perspective coming from and intertwining with active inference directions and robotics like we were talking about today.

JF.

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uh navigation and and and avoidance but that will not be clear so i think uh in that regard there there are gains but they're not they're not absolutely clear in terms of explainability but when it comes to active inference i think that what matters to me is that the robots behave according to the principle of active inference and not necessarily by implementing the mathematics of it um so i don't think it's a contradiction uh as far as i'm concerned these are four decisions that are our mind just symbolic programming and that's why i'm doing it that way thank you it's almost like in the textbook of active inference where we have the high and the low road and here's like the symbolic airdrop actually arriving at a similar place although being quite disjoint in its origins from the particular partition flows on continuous numerical variables a lot of the ways that we've seen things develop in the past several years so welcome carl if you would like to say hello or provide any remarks on anything that you've seen at the symposium or any overview thoughts on active inference and robotics well i've greatly enjoyed all the presentations there's so many different perspectives so um i will im i would imagine my comments will fall on from what people want to talk about so i'm going to just pick up on that last theme so you know the jean francois um symbolic dropping on from high i i thought it was a really intriguing perspective um and it's a perspective that i've been actually forced into in certain applications of the uh golo fashioned mathematics of active infants and so i thought it'd be useful to share that um um in the sense that when you write down um discrete

state space models and use the maths of uh minimizing variational expected free energy to invert those models very often you're confronted um with specifying those models with boolean logic you know if this then you're going to be able to learn that um and um you know literally in the code which you write to specify a particular tensor that maps from some some causes to some consequences you are literally using if and then

statements and and and or statements in order to generate the tensors or or or the um um the matrices that sort of uh encode probabilistic mappings which now become um um deterministic mappings simply because you're populating these tensors with zeros zeros zeros and ones so i do see an enormous uh opportunity here to actually um bring together the um you know the the mathematics and the simple probability theory just using you know um categorical distributions or Dirichlet distributions of a particular sort you know where you know you're only dealing with some zeros and one probabilities and the kind of logic uh and perception that you were talking about i thought that was absolutely fascinating and one one thing that intrigues me is if you can do that um then that that there is um a way of doing your searches of all the potential programs or logical um statements that are internally consistent um with the evidence at hand you know for example i could build a gerative model where i can be in two places at the same time or two things can be in the same place at any one time that's you know from the point of the gerative model i can write that down without an exclusion principle um but when i now come to sit and ask the question is that model relative to another model that can be written down in terms of boolean logic um that precludes two things being in the same place any one time does that simpler model now provide a more a better explanation for the evidence at hand um so technically what that means is that i can apply the rules of basic model selection to this model versus that model where the model just is a statement of propositions you know logical propositions um and there are ways of um there are ways of sort of automating that very efficiently called bayesian model reduction um and i'm just wondering whether you could take a lot of pressure off the search for the internally consistent logical structures that you're dealing with um if you can cast it in terms of um a um in terms of bayesian model reduction which quite simply is measuring the evidence the probability of this sequence of outcomes that can itself be categorical you know i'm in this state or that state um the probability of that particular sequence of discrete outcomes given this structure this gerative model that is articulated as a logic and if you can do that i think that'd be very very powerful because that speaks to a common theme which i think most of us or you were talking about which is this issue of structure learning um you know and what you're talking about is is sort of a principled way to get the right parsimonious structures that have this you know this coherence i can't remember you um you were even taking it right back to sort of 19th century philosophy or 18th century philosophy in terms of

you know constraints would be really great to see whether that kind of um

sort of logical structure uh the philosophers like um actually could be discovered or disclosed just using good old-fashioned basic model selection that there seems to be a close homology there with some of the work of

people like josh tenenbaum where he's he's he certainly articulates structural learning from the point of view of

radical constructivism using sort of program searches and sort of um um you know casting generative models as programs and then trying to score the quality of these programs uh using a form of basing model selection and he claims to be able to do sort of millions um very very effectively within within um

um yeah within a few seconds have you have you looked at josh tenenbaum's work at all

not yet i certainly will and i i just want to uh make clear that the app perception engine uh was

developed by richard evans uh in the paper making sense of sensory input um and and yes one of the the

the the the the core issues is inductive bias uh how do you restrict the search space so that you can

have a chance of finding within a reasonable amount of time a logic program that is a competent predictor

and uh those and that's where a constant uh synthetic unity of our perception uh plays in as it provides

constraints on the the class of uh logic programs which are uh acceptable uh which are are yeah which

are uh considered valid and if i make a contribution because this is work in progress

i would like to add additional constraints of from the fact that we do not uh we're not trying to generate

one generative model in in in of itself but as in the context of a society of generative models so a

general model will um borrow uh its perception domain from the belief domain of other generative models

thus creating additional constraints and as it searches it will try not to modify its scope in such a

way that it orphans other generative models so as the uh the society of general models grows

so does the set of constraints that apply on reducing the search for a a predictor in any of the

generative models themselves so i'm hoping that as the society of general models grows so does the set

of constraints that apply on the searches for any one of them but i think the name of the game is very much

in this uh the richness and the strength of the inductive biases that are applied on the search

just to highlight a few pieces of what were addressed and then to bruno or adam or anyone who'd like to add

it impressed on me robotics is radical constructivism not just in the sense of the constructing models but

also in the way that adam described in terms of the relationship to the embodied artifact so that is

one interesting piece and another in this quest to find the rainbow that connects first principles and

analytic formulations through mesoscale for example computational code all the way to the last mile in the

the cases that we heard from one hot trend in the previous session carl when you described the way that

boolean

logic enters at the level of execution was very interesting and suggests that there is kind of like a uh a

brackish area similar to that between discrete and continuous time models in like hierarchical modeling

but here where the numerical and the statistical approaches to active inference are becoming connected to

actual logical implementations and design choices that have to do with which functions are executed first

or the conditions that certain things occur in and so there actually is a meeting ground or an interface

in a way implicitly already and so it is very interesting to see how jf's work on the generative model

generation process is now approaching that intersection from the other side and suggesting it's like if you

see a car coming from the other direction there's a road and now this high and low road

network starts to grow more pathways so bruno or adam would be happy to hear any thoughts you have

um yeah thanks um yeah i mean i think that's that's um that's much our case as i as i was telling you like we come from this other side of of the story and at the end we have to make some choices on implementations and so far we've been very focused on on this um self-organizing maps most of our work is on that uh and that's actually one one of the questions i have for for all of you like uh what's your take on on all these approaches that we're trying to like we're coming from a different field from a different uh point of view and in a way we are almost uh intentionally avoiding uh probabilistic approaches to active inference no so i don't know what's what's your opinion on that that was the first one and the second one is more like a um i don't know like a a doubt a question for car like what do you see in the near future on on all these things so that we're working on on prediction error dynamics and this monitoring of the of the error over time um can yeah i think that's that's i'd love to speak to that if i can yeah and in fact interestingly does that also speak to your question daniel i just wanted to point out also that this notion of moving between continuous state space models and discretized state space models is something that people in quantum computing and physics are going to keep their close eye on because you get it all digital um then the opportunity for quantum computing that suddenly rears it rears its head so i think it was also a very pragmatic um pragmatic reason to sort of look at that um that transition and also just to bring in adam's point you know not only um was he suggesting that we quantize discretize spatialize in terms of little tiles and places uh receptive fields for being here as opposed to being over there but he was also suggesting we do that in time so i think the quantization of space time in a generative model is a really important thing of course you know you could argue that that's a step closer to good old-fashioned symbolic ai that has a meaning in in the sort of folk psychology sense so to come come to bruno's question so the big divide here is between um from my point of view at least degenerative models that are of continuous states and time versus generative models that are of discrete states and um states and time and clearly you need both you know in the sense that the world that a roboticist and indeed me and my children live in is a continuous world i have to move around it i have to um control continuous temperatures and all sorts of things and and yet it seems as if the the intelligence is at a symbolic level so i think what we're talking about is a hybrid generative model uh and then the question is how do you get the sort of discrete parts to talk to the continuous parts but if we just focus on the continuous parts um and just to answer bruno's question directly um first of all what i'm saying is you can't ignore the continuous bits so everything you would talk about in terms of self-organizing maps has to be there at some level um is this a question whether you can put a more symbolic um logical structure on top of it to core screen it sufficiently to resolve all the the complexity um the i i loved you your work um i could see uh all the important um issues in terms of um representing uncertainty as the if you like the stand-in for the valence of how i am doing so i was amused to hear that you're trying you're actively avoiding probability theory yet for me everything you said was all about probability it's all about uncertainty and that was true at so many levels so just to answer daniel's question to you uh the relationship between free and variational free energy and the prediction error is really trivial and very simple it's just under continuous state space

models with gaussian random fluctuations the free energy gradients just are the prediction errors the sign of prediction errors so it's really your simple quadratic forms for the log probabilities um so um self-organizing maps that organize themselves in terms of responding to prediction errors just are systems that are performing a gradient flow on a variational free energy under the assumption you're dealing with continuous states that have gaussian by random by central limit theorem uh gaussian random fluctuations so i you know that that's all exactly what you know uh what would please me if i was if i was a probabilistic committed to the probabilistic parts of it yeah you're i think your thing you're bringing to the table though is is is this sort of the dynamics of the uncertainty or the amplitude of the prediction error the unsigned prediction error the sort of the behavior of prediction errors over you know with the separation of time scales i think that that that's a really important um really important move um i i see slight homologues of that not formally identical but certainly um homologues of that um um from the point of view of an engineer this would be like getting the kalman gain right you know if you interpret a kalman gain on a kalman filter as assigning the right precision or confidence to the prediction errors as they come in relative to to the state estimators and your your prior beliefs that you've accumulated getting that right having an adaptive kalman filter where you're actually optimizing the kalman gain uh you know that would be if you like the state of the art hierarchical kalman-bucy filtering so that if i was a psychiatrist i would say this is exactly what goes wrong in people with uh schizophrenia and autism is that they've got their estimators of the overall um uncertainty or prediction error wrong um and that they now need to um you know and that this is an explanation for um false belief updating or false state estimation from the kalman filtering point of view the the other point um that struck me in the early part of your talk was um something which um as a neurobiologist or motor control theorist um something that they would find very entertaining which was the the link between the ability to ignore stuff and sensory attenuation i don't know if you've come across that in robotics but um certainly in terms of motor control of the kind i think you're using which is which i would um ascribe to the equilibrium point hypothesis like approach you know so you're you're putting your set point into into the actuator and you're letting the reflexes do the rest but uh you've got to control carefully um how those prediction errors are used to drive the actuator make sure they don't come back up and change your state estimation and so from the point of view of

sensory attenuation that just is attenuating the precision or the inverse variance or estimation of the overall amplitude so and finally you know looking at the long-term trends in the estimated precision i think is you know it is a really important way to see whether you've got the right generative model for this context um i noticed you were you're citing the work of um um matches uh joffoli uh you know and that was i think is really great insight many years ago interesting from economics he was working with economists uh uh at that time so so uh i think that's very very important as well it's all about estimating uncertainty yeah thanks a lot i mean i mean as as you can see there is uh well obviously we come from this this side of uh woolpert and you you can you can see all that all all that literature behind our work and uh but yeah we are getting there and uh that's that's uh that that's where we are going that's what

we're intending and you'll see some some work that we're going to come out uh very soon that tries to to bring deeper this this this um to make more useful um make better use of this of this um monitoring and the other thing i was telling you i was thinking is something you mentioned is this weighting of the attenuation and and that's always been in our in our mind again because of uh why can you tickle yourself and all these famous famous works um we have lots of doubts we have uh lots of discussion with with with people about this attenuation and and the precision weighting and that's what we are now trying to to implement um again using self-organizing maps but uh yeah we we will try to get results quite soon and thanks a lot yeah just just to say if you could i think that's really important because as soon as you can get dynamic dynamic attenuation or gating um in play very much in the spirit of what was implicit in um matt's um sort of homeostatic architecture you can talk you i think genuinely you can talk about your your robots attending to this or ignoring that in the right kind of way and you're starting i think to get quite close to now sentience because you've got attention in the mix uh you've got some a kind of metacognition in your self-organizing maps which i think licenses something which is much much closer now to to sort of a sentient intelligence um as opposed to just a reflexive kind of uh control yeah that's that's that's that's the idea let's let's hope for the best so a few points and then adam or jacob or anyone um active inference is inference about action and it also uses action as one of the ways to reduce uncertainty and so it's a very tantalizing parallel with quantum active inference as being about quantum systems and potentially using quantum in some way to run these models another point was in the recent folk psychology paper with with ryan smith at all there was a distinction of how increasingly cognitive and symbolic functions were being played by decision making active inference in the discrete setting while more motor behaviors could be played out by motor active inference in the continuous time setting and through this conversation i'm seeing how whether that is explicit or implicit that decision ai to motor ai handoff is mediated by a symbolic layer either implicitly through the code construction or potentially even explicitly with the kinds of approaches that we're exploring and then one last area of some parallels that i'll like to hear from adam or anyone else self-organizing maps brought up by bruno have a lot of similarities with some of the graph operations that tim for bellin and was on katal and adam's work was describing and then the harmonic modes that adam has described in work as well reminded me of the ultra stability that matt brown described in some of these more classical models of like synergetics and multi-scale harmonic organization and so adam or anyone else would be happy to hear any of those thoughts um okay just uh quickly to to answer adam um i don't know exactly how i mean what do you mean with oh but they can't see my question uh the audience maybe we should read ah sorry uh adam is asking to what extent can self-organizing maps be used as model of experience dependent plasticity as implementing a kind of implicit neural architecture search i mean i don't know exactly what you mean with neural architecture search but um um there is this this um literature on growing self-organizing maps so like the usual uh original architecture is is fixed but there is there is some implementations where you actually can add nodes and and they become more more well they have more plasticity and there is also

the the the ones that don't stop learning so it's it's continuously moving so that helps you to not have a fixed structure and a fixed mapping of the sensory input but it's it just moves as there is new data coming new input uh uh coming in so that's that's the dynamical sums and growing sums i think if if you are interested on that uh extremely uh i guess the the intuition or question was that um it's like yeah i'm sure there are plenty of inductive biases that we i'm equipped with to help us converge on efficient regimes of inference and learning but i'm also wondering the extent to which um i don't want to go too far with this kind of analogy but something kind of like a field programmable gate array but like um the that self-organizing map like dynamics could allow for um the creation of different regimes of active inference like like to what extent as you're like let's say um building up a hierarchy um um at different levels of it based on where you are in this overall hierarchy or heterarchy you might take on um very different computational properties um via experience kind of plasticity so it's like you you learn the inductive biases you need as go um but to come back around with self-organizing maps um the locality and the the um the topography of them uh one thing i'm really curious about is the extent to which they could actually be used for um modeling like um entorhinal and hippocampal representations to what degree the different operations that are um used for constructing these maps would also um converge with um the the kinds of structural inference and structure learning that have been

described for the hippocampal system so um i don't know the answers to that but it seems like a potentially fruitful intersection that daniel suggested so yeah definitely yeah i mean i think in principle they they are inspired from the hippocampal um activity you know the the maps and yeah there's lots of proposals out there of different ways of moving them but uh yeah i think they they could be very useful for what you want so that's what i'm talking about that like a general question for me that i'm i've been wondering about is so i guess i'm suggesting that like you're getting these represent these um representations structure representations potentially being um afforded by these uh place fields of the peccable system um but one thing i'm wondering is so and that these would provide like equilibrium points like uh carl often will talk about the importance of structuring action and perception um by these discrete acts that seem to be unfolding at roughly theta frequencies and this seems to be like the time scale of coherent action selection for an organism of about our size that can do about the things we can do with the brain about this big all kind of brought in the same temporal register um but and so this idea that like we might be stepping along these representations in some ways like moving from like between these bump attractors um as somehow structuring cognition by some sort of representation i'm excited by that but i still don't have like a sense like uh for how far you go with that or ought to go like for instance so let's say we're talking about something like um uh this is actually something uh uh uh carlos uh um uh school to me about in the past which is like before you like uh think of the architectural principle think of the inactive uh situatedness of the system and so something like i wonder is like so you know i'm focusing on like maybe you're getting this sort of inductive bias of like the hippocampal enterrinal systems this high level controller but to what extent are we implicitly doing things like symbolic reasoning but via like modes of enactment

like to what degree like could you do something like like let's say we're talking about like
pearly and causation and you want to do something like a do operator you could have some sort of like
graph some sort of like flow of inference like um maybe even like within like like chain bump
attractors like the hippocampal system and uh actually another thing that carl actually brought
up recently was on bump attractors in the colliculus so maybe even like in terms like patterns of like
ocular motor motion is potentially like providing like another means of accessing such representations and
being governed by them uh but to to what extent it is um i don't know is it too neurocentric is it actually
like being realized more implicitly by the overall contextualized functioning and like you it's you know i'm
open questions for me so sure to carl and then jacob or anyone else who wants to talk
to me well can i can i tell adam off again or or not i don't think he needs telling off um yeah it's
interesting isn't it um i wonder whether that is the um the interface that daniel was referring to
um how do you get from a necessary continuous um sort of bayesian filter coupling with the world i mean
even with um carla and andy you know these things are moving continuous states basically so at some
point even if it's actually um uh physicalized yeah there is a continuous uh generative model and
self-organization in play um and i'm reading um adam's question as you know is this too neurocentric
to to worry about bumper tractors and the role of lateral inhibition for example in carving up receptive
fields that could be actually written down as just sort of basically a quantized representation
you know for example um receptive fields in v5 or motion sensitive areas can either be expressed as a
sort of continuous um preference for motion as a continuous uh you know function of um visual
velocity or you can say no this population is just encoding the probability that this particular
edge is moving at eight degrees per second um so you know if if adam you you were asking you know um
um are we being too neurocentric in over interpreting the neuronal dynamics that just basically do the
carving up into a series of um either tilings and receptive fields or place fields or in um the
context of sort of stable fixed points you know sort of um um thinking about hetero clinic channels as
you know one way of articulating a discrete set of ordinarily organized uh fixed points that have you
know that's the way that we should understand it i i don't know but i think you know it could certainly
if you can read them in both ways i think that then gives you license if you want to simulate and
reproduce in silica these kinds of computations it certainly allows you to replace hetero clinic
channels um cycles uh bumper tractors receptive fields of this kind with a discrete representation
and you know uh to me um you know practically that would be an important an important way to proceed
simply because that minimizes the complexity of the generative model they minimize the complexity you
maximize the uh the model evidence um so you know i don't know whether being neurocentric is um
as a good thing or a bad thing it you know interesting just think about so what is what
what is a self-organizing map um so when i was at school it was this basically um uh a a couple map
lattice with with lateral inhibition to to um to do this sort of uh carving up and segregate those various
populations um or localities from other localities so if that's the case again you've got this notion of
lateral inhibition will it take all like dynamics what is that well is that a statement of this
exclusion principle and i mean it beyond simply your the sort of exclusion principles i can't be in one

place at any one time but the fundamental exclusion that you get when dealing with discrete state spaces you know i've got to be in this state and i'm not in any other state and that means that being in this state means i have to inhibit being in any other state just to get a sum to one constraint so just by going um if you like discrete you're naturally inducing by physical architecture requires the kind of lateral inhibition that is one of the defining features of a self-organizing map so that might be a virtue uh or you know of being a bit neurocentric in the sense of you know um understanding evolution as working its way to a coarse-grained minimally complex discretized symbolic representation of the world and we just now read receptive fields in a slightly oh we over interpret them they're just you know basically encodings the world is likely to be in this state from my point of view can i just make one final point um which kept on recurring to me that there'll be people chasing you on on this i'm thinking about bruno and self-organizing maps and adam in the uh and actually matt as well with

his um that i think there's a slow realization that these kinds of architectures are the way to go that's coming out of machine learning when people are getting bored with that propagation so they're now all turning to local energy um based rules and they now realize that they get much better performance from a deep neural network if they just use a local energy based rule and what would that look like it would look exactly like the self-organized uh self-organizing maps all three we've been talking about it's just me my note trying to treat everybody else around me um uh you know um and doing so by minimizing my local variation for the energy or minimizing the prediction errors uh you know mathematically those being equivalent the you know the twist here is that the you know this can be applied to effectively uh you know overly expressive amorphous a completely connected neural network and it will still work with the right kind of pruning you will start to get to the sort of nice hierarchical structures that Bruno was showing uh or the you know with the right kind of temporal scheduling you'll get to the nice you know the nice uh tolman-eckenbaum machines but at the end of the day these are just um they're just sort of deep networks or even deep that is networks with a local um a local energy and those things i've now noticed certainly the university of oxford are now called predictive coding so there's no necessarily no prediction there's no there's no prediction in the temporal sense that adam was trying to emphasize or in the sense of a calman filter it's just that they rely upon the local computation of prediction errors to generate the local energy function so i see you know the people the reactionaries in deep learning you know the young turks who want to do it better and they've all now identified predictive coding as the rhetoric in order to say this is how it's done and it's just a statement of of you know you don't need to do um i think bruno had a nice phrase do you do or perhaps it was perhaps it was um jean-francois defuse your learning everywhere through back propagation you if it's working properly you should just be able to do it locally with your own little node vertex of your som or gm in a uh in in jean-francois uh um if i could uh build on that and then anyone else who'd like to add a challenge that exists and is recognized in robotics and many of the presentations brought up was how to move from continuous and high dimensional data like a high resolution video camera or multi-sensor integration in the world

and then we're going to move from the world eventually towards logical and conditional contextual action and so when we see the neural network representations we kind of see sometimes zooming in and zooming out and different ways that people are training those models including limitations in those models training like the back propagation that carl just mentioned as well as that steven grossberg has gone into and so trying to contextualize that issue in light of what we're discussing about discrete time and about symbolic logic it seems that through the partially observable framework not even saying the partially observable markovian framework but here including holographic principle and so on there's a way to move from a continuous statistical distribution and make a map to a discrete statistical distribution like the a matrix in the pomdps that we use can have a continuous variable and then map that onto a discrete statistical distribution and then i wondered if there's something like a logical a another tail of two densities a tail of two different cities or a third city and so this matrix recognizes logic from discrete distributions that's the recognition density and then the generative density is the emission of logics that are compatible with discrete distributions because with the recognition direction recognizing conditional logics and searching over relatively small sets of possible conditional logics from discrete distributions is possible like a verb is never used two verbs in a row are not used or something like that from a discrete distribution logic could be extracted or recognized and then if there's a discrete distribution or more logics can be sampled and that speaks to that inductive bias

where you want the inductive bias to have limited false positives and false negatives with respect to the empirical world

like you wouldn't want to waste time spitting out logics that cannot be however that might not be lethal but it certainly wastes a lot of time

and then conversely what could be lethal would be to have an inductive bias that fails to recognize empirical aspects of the world

and so that is like an expectation maximization but transposing not between a continuous and a discrete statistical space

but rather between a statistical and a logical space

so yakub or oh yes yakub or bruno anyone who would like to add to that please

no yakub uh you you haven't said much uh please but yeah i i can say something afterwards

um yeah i just had some general uh questions that are kind of related to um daniel's remark on the move between the purely continuous and statistical representation of degenerative molecules into the discrete and um since this week i've been reading the paper on from chris fields and carl um and others on the fep and neuromorphic development i'm wondering whether the question of structure learning can be solved by some fundamental computations that are available at the very low um low dimensional level of just quantum quantum systems where given given the right um environment that system would naturally evolve into some kind of discrete morphology that it didn't start with it kind of similar to like the primordial soup then evolving into discrete um or more discrete life forms um and whether whether and whether that might impose the the necessary um structural learning principles where or um message passing where each um each new layer that evolves um in the hierarchy is receiving observations

from

receiving observations from the other ones um given by the new markov blanket that's drawn around it um but in terms of actual implementation i uh

uh i'm wondering whether that requires quantum computers or whether that can be done with some kind of different

kind of different architecture that is able of this uh automatic um uh evolution in its morphology

awesome bruno and then anyone else who'd like to add

um yeah very i mean very nice your your your question and then um just kind of to round up what what you were all

all concerned about um we were just thinking and remembering this this and going back sums and there is this uh proposal some sums but i don't remember the authors now but it's this other type of architecture that takes into consideration

um the the learning progress so previous activations and some sort of history of of the winning previous winning uh nodes and so on um that that might be the next step so that you can actually monitor what's happening over time

and if you have one of these that it's coding for error then that would include the monitoring of of the prediction error yeah and and it would kind of close this gap between continuous and discrete representations because then you could you could have it all over time so i like if you wonder that i can send you some some literature on that that

that

thank you gruno one point on the morphological computational angle uh jacob and then again anyone can raise their hand

i thought about a bunch of tree seeds we're back in frequentism again so everyone can take a breath but you know there's a hundred seeds that are planted

and depending on the priors inherited from evolution and the updates which are happening from the generative process from the niche

there's going to be like a distribution of that tree through time so like a palm tree is going to have a very like narrow cone some sort of distributional cone across those hundred

that are sampling from it and then another a shrub might have like a different um shape

then you brought up how potentially quantum or discrete like morphological decisions could realize that continuous probability

distribution at the population or even like the multiple worlds level and there's something there also with the local computation of large models

that carl mentioned that it's like something happens where there is not even just a symbolic or discrete decision that's made

there's an embedded decision that's being made that now is part of the history and so the branching pattern of any given palm tree is going to be unique

unique or any given shrub yet they also may fill out at the population level a distribution set that has these aspects that can be modeled as like a gaussian blur over tree morphologies yet of course no one is saying that the tree is a blur and so there's so many interesting contrasts with like the realized trajectory of the lego robot

the lego robot and that's the n equals one the population of trajectories the imagined set of trajectories so having a unified ontology to be able to talk and have formal concise connections amongst these different kinds of remembered now casted anticipated or imagined futures helps find the patterns across systems that like adam said are essentially the basis of sense making and insight and insight

i gotta ramble um so i guess in terms of this like um motion from a continuous to a discrete regime or even like um drawing analogies from like a quantum to like a more classical regime uh

uh there's a general like i've been kind of like with respect to computational models of consciousness wondering um how um a seemingly classical world of experience can emerge from a probabilistic model like why are things so precise and or um carl's written some really uh interesting papers with andy clark on this um on the bayesian blur problem and one of the suggestions was that

was that like the discretization for the sake of action.

Like in order to act, you have to act in a particular way.

And it's like just the requirements of doing a particular thing at a particular time induces this.

And so one of the things, and so this is,

I don't think it's competing,

but in some ways competes for like a suggestion I had, although I think it's compatible.

Cause like I was wondering whether basically like something about the timescales at which you could get like these coherent eigenmodes or like the population activity could achieve these like attracting states would create like the timescales of the formation of these large scale attractors could create like an in and out of like who gets to inform or not.

And this could help to like sharpen things up and like creating these population level attractors.

But there's another way of describing it in terms of like what lets you act.

So in terms of like these different,

I guess one more comment along those lines is,

I think it's interesting that like,

so I've talked to some rodent researchers

and you,

when you remove the hippocampus,

you'll still actually get like roughly theta scale,

theta scale,

like large mesoscale organization of brain dynamics

for the rest of the rodent's nervous system.
So it seems like there's almost a,
it might be,
I don't know if you'd say it's like degenerate,
but it's like natural selection,
like any mode,
any,
any way it could catalyze things to help with the coordination,
to help with the alignment,
the spatial temporal alignment of,
that,
that allow the call out for different forms of synchronization
was utilized.
So like there's like,
you could get,
for instance,
a good amount of discretization just from like,
potentially just from like learning and interacting with the world
of that requires you to do this just physically to interface with it.
You get some of it from like the overall connectomic properties,
just like that's the times,
like that's,
what's required for the,
the,
the,
you know,
the overall brain to form attractors that they'll form it at a certain
time scale and in different nervous systems of different size and
complexity with different bottle,
like information bottlenecks might tend to form these attracting states
roughly on that same scale.
And maybe things are tuned for that,
or maybe they get tuned by experience,
either as an innate inductive bias or as a empirical,
meta prior learning or learned.
You can also get it with things like the local logic of like,
how you transition between these different equilibrium points for like
highly central structures.

And so something that's very unclear to me is like the extent I imagine the answer is like all of the above and like most combinations, but it's like, to what degree were these inductive biases, are they taking the form of evolutionary priors and to what degree are they developmental priors?

This is still very unclear to me.

But one thing that, and there seems to be a tension in machine learning in terms of these, like you want these rich inductive priors to do efficient inference and learning, because otherwise it's hopeless, you pay a price in terms of generalization sometimes.

And so I have no idea how nature balances to what extent and what cases, or I have some idea, but not nearly enough to feel comfortable.

So that's my ramble.

I think we can actually hinge on this and bring it to an area that some of the registrants brought up and also something probably many of us have been thinking about, which is multi-agent modeling.

And I wanted to connect that to what Adam was just saying about what is granted via embeddedness, what arises or becomes more possible simply as a function of realized corporal embodiment, as opposed to in silico simulation.

And in our multi-agent discussions, we've been differentiating spatial multi-agent scenarios from essentially non-spatial, like digital or cognitive.

So in the spatial multi-agent case, the real world, the niche,

the generative process does the work of inducing or preventing collisions.

Like it just cannot be the case that two entities are in the same location.

Whereas in the cognitive case, whether it's birdsong or a negotiation or verbal communication, or we're visiting websites, we can both be at the same website.

And so then there's sort of a mass parallel stigmagy occurring or a mass parallel real-time

architecture where there is some kind of coordination,
but the coordination isn't exclusionary.

Again,

because two entities could,
their thought trains could essentially cross in and out.

And those trains are like ghost trains that don't exclude each other.

Whereas that couldn't happen in the real world.

So I wanted to ask anyone who had thoughts,
how does multi-agent modeling come into play for robotics?

What current issues are facing multi-agent robotics?

And how can what we're discussing here with active inference play a role specifically
in understanding complex multi-agent scenarios?

I have a question in that regard for Matt.

Matt, your solution has been applied to a single agent.

Do you see the possibility of a homeostat that is composed of multiple agents
that will direct their collaboration towards achieving a common goal?

Does it scale to multi-agent situations?

Thanks.

I hope Matt can provide a thought as is working.

Carl and then anyone else.

Where is Matt?

Listening.

Can he speak?

Only at a future point in an expected trajectory.

I will answer for him then.

I would imagine what he would say is if you just take the generalized synchrony perspective
on the emergent properties of a homeostat coupled to its environment

and you couple two homeostats together,

they will find a mutual homeostasis that just will be a joint synchronization manifold.

So what you'd expect to see is a coming together exactly in the spirit of dynamical generalized
synchronization.

So there will be, as I think Daniel mentioned earlier on,

the homeostats will be singing from the same hymn sheet under the constraints of what they can
communicate.

So I think from Matt's point of view,

he's more interested in establishing a generalized synchrony between a use case,

an industrial use case,

and his homeostat,

as opposed to just letting two homeostats shape and design their own little eco-niche,
and a little bit of a connection indulgence,
sort of cultural niche construction.

While I'm talking though,

I think there's a nice,

something very interesting about Daniel's question in relation to your work.

I sort of overheard the little robots telling each other what they were thinking.

And I thought that was really important.

And simply because,

you know,

speaking to this general question about,

you know,

what does active inference or what,

you know,

what are the special considerations that you might want to bring to the table when thinking about a multi-agent setting?

I think there are two ways we could talk for ages about this.

First of all,

it's having the law of requisite variety across agents,

and then the story would unfold in terms of natural selections based in model selection,

and that would,

I think,

answer large parts of Adam's questions about where the inductive biases come from.

If you cast the biases as selection biases,

we're just talking about Bayesian model selection,

selection amongst what the requisite varieties afforded by Ashby's law.

But to make that Bayesian model selection work,

you have to have natural selection.

So,

you know,

there's a great story about the importance of multiple agents from that point of view,

from the point of view of structural learning for free as an emergent property of natural Bayesian model selection or natural structural learning.

But the other thing is,

just from the point of view of active inference,

if you've got multiple agents,

what's to stop you thinking about these multiple agents as one big agent where you've cut some message passing between them?

And like one big agent with lots and lots of eyes and sensors and the like.

So I'm asking now,

when I looked at Carla and Andy,

what's to stop me thinking about Carla and Andy as one agent that has a really flexible and deployable set of sensors?

So they've got eyes that can point in different directions.

So if we translate this into sort of,

you know,

controlling drones,

for example,

and we have a swarm of drones,

we can think about that as one agent,

one robot with lots of deployable eyes,

which gives you,

you know,

an enormous flexibility over the,

you know,

possibly complexity of the way that we control our two eyes,

which are both in front of our heads.

So how would you then write down a good generative model for a swarm of eyes where there's one CPU, there's one,

there's one sort of,

I can't remember,

the Lego brick controlling all the drones.

Well,

you just write a generative model with multiple sensory modalities where you needed to deploy action in the right kind of way to get the right epistemic foraging or whatever.

But that would entail now message passing,

belief updating,

natural interactions,

an SOM,

between the brains of agents.

So say,

no,

you can't do that.

You've actually got to have physically separable robots to have a truly multi-agent setup.

So how can you now work around the fact you don't have direct message passing and belief updating between the brains,

the brain,
the separated brains,
where you just have communication.
So provided you broadcast what you believe,
and provided those beliefs are perspective independent,
so they're conserved from,
you know,
so that they,
you know,
what I say in terms of my frame of reference is meaningful from your point of frame of reference,
which,
which,
incidentally,
for Daniel and Jakob,
becomes relevant from the point of view of quantum frames of reference here.
But if we can assume we've all got the same,
we've all got an allocentric frame of reference,
then all of it is required to put together the many brains into one brain,
is to have beliefs broadcast,
just those beliefs that will,
that will,

that are conserved or shared in the,
in the ensemble's derivative model.
So,
you know,
I don't know where your,
was Carla able to hear Andy and vice versa?
So what did they actually share a narrative of belief updating?
No,
they did not.
They,
they,
the,
the,
the speaking was,
the talking was mostly for my benefits.
I could see what was going in their mind at the time.

So when,
when Carl was saying,
oh,
Andy is freaking out.
I knew that the,
the generative model had identified the behavior and,
and,
and inferred the mindset of the other robot.
But I,
I,
I am playing with the idea that they can communicate by just distributed message,
passing from one robot to the other.
And what they would communicate is,
is essentially,
it,
it,
it,
I would,
I would consider one agent as,
as a,
a source of sensation for another agent.
And,

and in the same way that internally,
generative models,
uh,
the beliefs of a general model becomes the sensation of,
of,
of another higher level generative model that can scale across multiple agents.
And,
and that would happen by communicating,
uh,
these,

these,

predictions across agents and this prediction errors across agent,
as they propagate between general models within single agent,

and just scale the architecture.

architecture.

So instead of having a society of mind within an agent,
which would also have a society of minds across agent and the same mechanisms would be at play,
but you're absolutely right.

We would need to have this communication,
but it would be the same kind of communication within,
uh,
across society of minds as they would be within the society of mind.

I'm,
I'm hoping that it would be self-similar and the same architecture would apply at any scale.
which actually speaks to Jacob's question,
uh,

you know,
Markov blankets and Markov blankets and sort of,
you know,
that sparse do you get from,
from that scale free separation and,
you know,
refined,
minimally complex message passing.
I think there's something quite fundamental there about,
you know,
the nature of communities and,
and the,
you know,
the scale of the world in which we live and all the organs that comprise me living in that world.
That,

that scale free aspect that is defined by the sparsity and the absence of coupling and,
and getting the messages that are communicated,
right.

I think that,
that's something quite fundamental about that.

I got a question.

Adam,
please.

Um,
so for the internal,
uh,
organism internal case,
I'm wondering,
so to what degree do you get,
um,
common agency for,
um,
societies of minds due to common embodiment?

But when we're going for a kind of,
um,
common,

or,
or joint agency,
and maybe even different forms of joint identification or,
or D individuation,
um,
or re individuation of individuals into a collective?

Um,
to what degree do we need something like,
um,
as constraining as a joint embodiment?

Like for instance,
like I'm thinking of like,
um,
uh,
soldiers marking and like marching in step or like the ways in which like,
we kind of like seem to like,
potentially even like expand and hack our body maps by like sharing like,
different synchronous modes.

You know,
that literature is somewhat contested.

Um,
I like the literature,
but,
um,

the,
but to what degree is it,
you don't need to have that kind of synchrony though.
It's just like a strong enough shared task is enough to like,
um,
radically make someone,
uh,
uh,
make the central thing,
their role of where they're singing in the hymn sheet,
that becomes the primary attractor governing them.
And you don't need to like,
um,
necessarily have this,
this kind of,
uh,
physicality.
I don't know if that makes sense,

I'm thinking of the different means in which you,
yes,
do you need something like synchronous in time,
or is it just a very strong,
um,
pressure,
selective pressure for coordination to establish a synchronization manifold to create,
to pull off the re individuation.
Um,
if I can't get any put chat in mind,
but yeah.
Oh,
it's a great question,

So thank you.
I also think this ties to again,
JS framing of society of mind,
and then we can contrast that with a society of bodies.
So the society of mind is that virtualized case,

whether it's the counterfactual virtualizations that cognitive entities can be modeled as doing,
or whether it might even be a society of minds in a digital setting.

And then there's the societies of bodies,
which maps earlier to our multi-agent discussion on like a physical crowd.

So like,

what are the similarities and differences with a digital swarm and a physical one?

It comes down a lot to their ability to collide.

And again,

the digital swarm can weave in and out because they can coexist in the same location in a semantic space in that generalized foraging way.

Whereas the embodied swarm is going to have collision prevention by virtue of the physicality.

And I think by using active inference to study these systems and integrate them,

we do approach exactly what you brought up, Adam,

which is like almost what do we get through shared task as opposed to what do we get from a common task?

Something that we merely have in common versus something that we're actively coordinating on together

and potentially even coordinating together in the same exact space or on the same instance of,

which provides the most constraints.

Like if people are rowing the same boat versus rowing parallel boats versus on different lakes.

So as we sort of separate spatially and especially virtualize,

there becomes more and more possibilities.

And this for those who are coming for perhaps from a robotics angle and wanting to understand what active inference brings to the picture here.

Hopefully we've pointed a few things like the scale free or scale friendly nature of models to be composed based upon their sparsity

and defined interfaces and also the multimodal aspect.

For example, Carl asked JF if the audio could be heard rather than only emitted.

And that's not going to require some bolted on communication module.

It is going to be able in the symbolic or in the numerical case to be mapped onto an architecture that also Bruno showed.

These architectures that can do sensor integration as a function of the interface definitions locally,

rather than potentially have ad hoc structural design decisions that could result in a lot of research and engineering debt.

So happy to hear any thoughts on that.

Thank you.

I'm.

I'm.

Yeah.

I mean, yeah, you're right.

I'm.

No, I just wanted to add that we are working on that community.

Well, first.

I come from from the very I don't know, like a very.

like a very now you could call it old-fashioned school where we were we as you can see in our implementations they are kind of very low level what we could say low level and and coming from the sensory interaction with the world and that that is maybe why we are not or we haven't been so far very concerned with very high level representations known so we are just dealing with very basic interaction with the world and at the same time it's it's my belief i mean it's very interesting this interaction between agents and so on but i think or we i've always thought that first we need an agent that that knows itself in the world and and has an uh its own model so like a first frontier or a first blanket as you like to call them um and only then can start to understand other agents and to to interact with other agents um yeah so that that was what i wanted to tell you know thyself

carl i'm just wondering there is an argument from um developmental neuro robotics then to have a true sense of self you you've got to have a true sense of other and you can only ever have a true sense of other if there's something else out there that's physically very much like you so just i'm speaking now to adam's uh point you know does it have to be uh or perhaps daniel's question you know the the the shared body as opposed to the shared the shared narrative you there there is an argument uh i'm not making the argument i'm just saying that there could be an argument that in order to disambiguate the causes of some sensory consequences of an action from uh where the action could be made by you or an identical robot or me or mum or me or my brother um that is the only case in which you are now going to be able need to um contextualize and assign and attribute the agency of this outcome to what to self versus other in a world in which i was the only object or phenotype like me i wouldn't need a sense of self and i wouldn't need a sense of other it's just when you have multiple agents that are quite similar and confusable um then you actually need a sense of self to make sure it's not it's you and not me um which of course you know speaks to not only theory of mind but the other the essence of communication as well so i'm just making that point that you know um that putting two robots together maybe as you point out once you've sorted out a robot just learning how to move um then learning that that the consequences of it moving could actually be reproduced by somebody else and then it might develop a sense of selfhood or a minimal sense of selfhood and then i think you're in the game of true theory of mind that's uh yeah it's a very interesting area with thinking through other minds as have been brought up in some of the presentations and then society of mind it is that nesting of blankets of different kinds of generative models but again using the same statistical or formal machinery just like we could have a nested multi-level regression model where one model was in kilometers and then there was another model that was in degrees uh for temperature um and also the question about swarms and collectives and groups comes up in biology all the time with different ways that people

delimit or qualify individuality ranging from the evolutionary unit of replication the kantian concept of an organism as the unit that is teleologically closed the physiological individuality which might even include cases like symbiosis the information theory of individuality and different ways that uh we can look at information processing and transfer in terms of what individuals are and uh it's so fascinating how that comes into play with robotics where when we think about the robots that will be interacting with us as a slightly different set than the ones that might be working underwater on a pipeline they're like like matt showed they might be able to interact with a relatively simple generative process that doesn't exhibit mind-like qualities it's interacting with a mere active inference entity but what on the other side of the blanket there's another adaptive active inference entity and especially when it's us then there's a space of norms and also laws which are like that pseudo code execution order for thinking through other minds in a way how will robots navigate a space and how will they be able to make abduction occur in a real-time way both in the generation of novel hypotheses with inductive bias followed by the selection of trajectories of action again taking into account formal codified law as well as the kind of state of exception competition is known to make the competition even or understanding when preferences allow for something to be so strongly desired

it's pursued might not be how you'd pursue it in a different time.

So there's just a lot of cool threads.

Yes, Adam, anyone else?

This is reminding me also from the first session, there was mention of value alignment.

And this seems to be like a multi-scale problem in terms of like it will show up just as much of like the robot not stepping on your toes or you're crushing your foot to like as it expands, you know, does it, is it, is it optimizing in the direction?

Are its tensors going in the direction that you want your tensors to go shaping?

And it seems like of the proposals, sometimes there's, how do I say this?

Like begged questions that active inference may be answers.

So it's sort of like, let's say you have a framework like cooperative inverse reinforcement learning.

And so there's some sort of like, you know, so it's like you are trying to like roughly like optimize for the utility function of the other agent.

So how do you get this kind of like shared utility?

But then like these like active inference models where like you're bootstrapping minds like intersubjectively, you have like regimes of joint attention and thinking through other minds.

Or I mean, some of this work with them I'm doing with Anna Chianitsa, it's like she emphasizes how like we start out like actually physically inside of another organism, completely dependent, like

homeostatically.

And so like you're automatically sort of grandmothering in this joint intention right from the get-go. And the seeds of, you know, organization and individuation are already like, you know, scaffolded in that way, just by the, just by the, by the niche contextualization where you're co-constructing each other as your mutual, you know, you and me are intersubjective like mutual niche construction.

So it seems like with respect to, I don't know at what point these problems in terms of like deploying robots when they first center, because, you know, it's, it's on, you know, always highfalutin as like intelligence explosions, like a recursively amplifying system stay aligned with you to, but just like in the world, you know, how can I make sure the robot like doesn't crush my foot?

Like I don't know.

I know.

So the ways in which co-valuing and the robot identifying itself as an individual relative to other individuals plays into it, it seems like active inference has like a very rich way of handling and modeling a lot of these different proposals and processes.

But something I would wonder like for, to ask like actual roboticists, like how near term is it like a live problem of getting like what forms of social modeling for robots, like, and what forms of like a sense of self modeling for the robot, like how much do we need at what stage for what levels of deployment in the world? With what degrees of robustness?

Simple question.

Great question.

And maybe just to double on what active inference does to potentially frame that junction, since in this round table, we won't address, but this is the stuff and substance of applied active inference symposia for many years to come in an economic framework, a reward absolutist framework, which is kind of a full stack ranging from the ways that models are trained based upon pragmatic value and reward all the way up through a world that values economic returns.

The question becomes in human robot alignment or in multi-agent alignment, or generally the question becomes how are we creating and then allocating epistemic, I'm sorry, how are we creating and then allocating pragmatic value? And epistemic can be valued to the extent that through time it provides pragmatic value. For example, one person doing a little bit more of a research angle and one doing more of an application and profit side of the business. And when we shift to a uncertainty imperative for a sustainable or even optimistic world model, then we can achieve pragmatic value as a consequence of a process theory that highlights reduction of uncertainty. And then the question moves from creating and allocating value to how can this ecosystem of diverse entities find a general synchrony and communicate and be how they are in a way that will be reducing the expected free energy of the ensemble, which is not the question that was even approached by the create value, allocate value framing that has come up in AI and continues to occur to this day. What if the image creators, what if they

end up reducing our food allocation so they can have more, they take the intellectual property instead of providing it to humans, that might be framed slightly differently in an uncertainty reduction framework plus optimism rather than in a reward maximization, which somewhat enforces a atomism and even a pessimism.

I just wanted to agree with you. That was an excellent point. Yeah. Yeah. I mean, there are so many ways you can tackle that.

And I was still puzzling over Adam's challenge. How do you stop robots standing on your foot?

I was thinking, how do you stop little puppy dogs weeing on the carpet?

Yeah, I think it's really you got to train them, haven't you really? I don't think there's any magic answer.

And I think the answer is just what Daniel said, you know, things are designed and can presumably only exist under the free energy principle.

If they minimize their uncertainty or expected surprise so that they make everything as predictable as possible,

which simply means that if I'm living in a world full of dogs, robots, cats, flies and plants,

then we're all trying to make ourselves as predictable as possible to each other.

And that is the, you know, that's a sort of folk psychological way of saying that

variation free energy is an extensive quantity.

So if a set of Markov blankets or phenotypes of varying forms and structures jointly has its own Markov blanket,

so it is an ensemble or a group or a community or a family, then it must put a Napa bound on its surprise and free energy.

And because the free energy of each constituent of that ensemble is summed to give the joint free energy, then it also requires that each individual is trying to minimize her surprise at the same time.

It's all internally consistent and quite, you know, to my mind, unmagical.

If you just look at it in terms of minimizing free energy or minimizing optimism, surprise under optimism, as Daniel nicely put it.

Thanks.

You just, you just have to make your dog surprised when he wheeze.

So you've got to find a way of making the robot surprised when he stands on your foot.

JF, you have a dog, right?

So how do you see similarities and differences with these versions of,

and also potentially even a family?

So how do you see these different entities being,

communicating and, and behaving appropriately or not and learning?

Well, that's having a dog, it raises the question of, do I make myself understood?

And that's a very, that's a very difficult question to answer.

When I say, no, you're not allowed to eat on that plate on the table.

I think my dog understands.

I'm not allowed to do this now.

As soon as I walk out, then permission is granted.

So I, you know, it's, it's, it's difficult to communication.

It's very difficult across, across individuals, especially across species.

But I'd like to, to just, uh, just rent a little bit on something which, uh, came up, which is, uh, the scale free nature of, uh, free energy principle.

The fact that it can apply at a societal level, the same way that it applies like a Russian doll to the cellular level.

As a software developer, I see self similarity, the recursion, uh, fractal as, as a, as a beautiful thing.

And I think at the same way physicists have symmetries as, as a guide to what is right, because it's beautiful.

I think we share, um, self similarity as, as a guide as what is beautiful and possibly true.

And I think self similarity, um, is beautiful because it's, it's, it's, it's, it's, it's collapses the amount of information that you need to generate something rich.

Um, so that's why I think that having, um, a society of mind made up of, of actors and having a society of minds, uh, and having the same principles scale up, uh, feels very beautiful and feels right.

Excellent point.

Um, yes, just if we could go for any, any length of time, a final set of thoughts and reflections from each of you.

Quick interjection, uh, Jeff, if you have anything written on that or want to write it, there's an upcoming special issue on symmetries that, uh, I think it might fit with.

Would you like to give any final reflections or anything that you've heard or felt updated by any thoughts you have on the state of the applied active inference in 2022 in robotics or where we're heading now that this symposium is behind us?

I'll start with Adam, but then everyone else can go.

Um, thank you, Daniel.

Um,

well, this simulated a lot of questions and just seeing the range of work being done was inspiring.

And, um, all I have to say is I was grateful to be a part of this.

Thank you for putting this together.

Thank you, Adam.

And after the closing words, I'll review every presentation and list the co-organizers as well.

So Bruno, and then we'll continue on.

Yeah.

Just, just like Adam said, thanks a lot for the invitation.

It was, it was very kind of you.

I think I want to thank, uh, Mark for insisting, Mark Miller that was insisting on us to, to present with you here.

Um, it was, it was a very nice experience and, uh, uh, well, we keep, we keep working.

We need more hands.

We have more ideas than hands, but I suppose that's everyone's problem.

No.

So yeah.

Thanks a lot.

Great.

Then there were three.

Uh huh.

Jeff.

Yes.

Well, again, thank you very much.

This was extremely stimulating.

Every talk I came up, I came out, um, listening to it with this.

I want to introduce in my robots, some aspect of it.

Uh, and, and my mind is, is, is buzzing right now with, with possibilities.

And so thank you so much for this opportunity.

Um, I guess, uh, I can go next.

Also, thanks a lot for, uh, um, being able to join this, this final round table and for all of the great presentations.

I'll definitely have to look over them, uh, probably a couple, couple more times, uh, to fully, um, understand all the progress that, um, has been, has been made and specific specifically on, there's a lot of, a lot to consider on the multi-agent, uh, modeling and the ideas, uh, in this discussion that, uh, we're, we were also talking about quite a lot in the active blog friends, uh, project, um, probably was left with, uh, more, more questions at the, at the end of it, but that's also the exciting part of it.

So thanks a lot.

And now it's my turn.

I cleverly leave myself to last because everything that needs to be said has already been said.

So that just means all I have to do is just to thank Daniel and his colleagues, which he's going to thank anyway, but personally to, to Daniel, thanks so much for all your energy, not just today, but over the years, it is really great to see these people and colleagues come together and sort of, you know, generate ideas and questions, which is exactly what we need to do.

Thank you.

Thanks a lot, Carl.

And I think my one personal reflection, it swooped in at the end when JF said that each of the presentations, there was like a module or an archetype or a function that transposed. And I think that is proof of concept of working in a shared ontology, as well as in an embodied

shared ontology, which is to say like a field in a community of practice with different small world structures and so on.

But when we have the ability to transpose across systems and scales, then we can bring in the modules and the pieces that people are mentioning and start to understand how those functions can be composed and designed.

And that's extremely exciting.

Everyone, thanks for joining the round table.

You can depart the Zoom or I'm just going to review the presenters again and the co-organizers.

So thank you for joining the round table.

You can leave if you'd like.

In the first session, there was Tim Schneider with Active Inference for Robotic Manipulation, Tim Verbellin, Robots Modeling the World from Pixels Using Deep Active Inference, Ben White, Artificial Empathy, Active Inference and Collective Intelligence, Norsajit, Learning Agent Preferences, and Wenhua Chen, Dual Control for Exploitation and Exploration and its Applications in Robotic Autonomous Search.

Then in the first round table, it was Wenhua and I talking about dual control.

The second session began with Bruno Lara with Prediction Error Dynamics, a proof of concept implementation, Matt Brown, Real-time Robotic Control through Embodied Homeostatic Feedback, and Adam Safran with Generalized Simultaneous Localization and Mapping, GSLAM, as Unification Framework for Natural and Artificial Intelligences towards Reverse Engineering the Hippocampal Enterrhinol System and Principles of High-Level Cognition.

And then JF Cloudier towards a symbolic implementation of Active Inference for LEGO Robots.

We've just completed the second round table with Adam, Bruno, JF, Carl, Jakob, and I.

The organizers for the symposium were Mark Miller, Matt Brown, Blue Knight, Alex Vyotkin, Yvonne Metelkin, and myself.

So thanks again.

Till number three of Applied Active Inference.

Hope to see everyone around the Institute.

Bye.

Bye.

Thank you.